MILITARY SPECIFICATION

TRUCK, TANK A/S32R-9

1. SCOPE

1.1 This specification covers a 5,000-gallon capacity fuel servicing tank truck, designated A/S32R-9.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-403	Plastic Molding Material, Polytetrafluoroethylene (TFE-Fluorocarbon)
O-A-548	Antifreeze, Ethylene Glycol, Inhibited
W-C-440	Clip, Electrical, Jaw-grip
VV-F-800	Fuel Oil, Diesel

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-W-1511	Wire Rope, Steel (Carbon) Flexible, Preformed
MIL-H-4441	Hose, Rubber, Gasoline, Lightweight
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-N-5877	Nozzle, Pressure Fuel Servicing, Locking, Type D-1, Nominal 2-1/2 Inch Diameter
MIL-M-6404	Meter, Volumetric, Positive Displacement, Liquid, Type F-1, Aircraft Fuel, 300 Gallons Per Minute
MIL-H-6615	Hose Assemblies, Rubber, Fuel and Water, with Reattachable Couplings, Low Temperature
MIL-E-7589	Extinguisher, Fire Vaporizing, Type D-2, Bromochloromethane, Portable (Extinguisher, Fire, Bromochloromethane, Portable, Type D-2)
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for

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MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-T-12459	Tire, Pneumatic, for Military Ground Vehicles
MIL-L-19537	Lacquer, Acrylic-Nitrocellulose Gloss (For Aircraft Use)
MIL-A-25896	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MIL-H-26521	Hose Assembly, Rubber, Fuel, Collapsible, Low Temperature, With Reattachable Couplings
MIL-N-26978	Nozzle, Fuel and Oil Servicing, Over-the-Wing, Shockproof, Type MD-1 and MD-2
MIL-C-27487	Coupling Halves, Quick-Disconnect-Cam-Locking Type
MIL-H-38209	Hose Reel Assembly, Fuel Servicing, General Requirements for
MIL-C-38479	Coupling Assembly, Quick Disconnect, Fuel GRK-15/S32R
MIL-T-45341	Trucks and Truck-Tractors, Commercial, Gasoline and Diesel Engine Driven, 34,500 to 64,000 Pounds, GVW, 6 x 4
MIL-F-52308	Filter Element, Fluid Pressure
MIL-R-83232	Reel, Static Discharge, Grounding RMU-20/E

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS24484	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MS27610	Connector, Electrical, Ground
MS35000	Battery, Storage, Lead-Acid, Waterproof, for Tactical Vehicles
MS35647	Padlock, Key Operated
MS51118	Pintle Assembly, Towing-40,000 Lbs Capacity, Manual Release

Air Force-Navy Aeronautical

AN2552	Receptacle - External Power, 28 Volt DC

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DRAWINGS

Air Force

50C24046	Coupling - Male, 6-Inch Air Duct
69B52267	Plug, Electrical, Ground

Corps of Engineers

B13201E6470	Plate, Seal, Intermediate

PUBLICATION

Air Force Bulletin

518	Cargo, Aircraft Compartment Dimensional Data

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers

SAE J695	Turning Ability and Off Tracking

(Application for copies should be addressed to the Society of Automotive Engineers Headquarters, Two Pennsylvania Plaza, New York, New York 10001.)

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

Department of Transportation

Motor Carrier Safety Regulations

(Application for copies of the above two publications should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

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American Society of Mechanical Engineers

ASME Boiler and Pressure Vessel Code
Section VIII Pressure Vessels
Section IX Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

American National Standards Institute, Inc

S1.4 General Purpose Sound Level Meters
S1.6 Preferred Frequencies for Acoustical Measurements
Z24.10 Octave-Band Filter Set for the Analysis of Noise and Other Sounds

(Application for copies should be addressed to the American National Standards Institute, Incorporated; 10 East 40th Street; New York, New York 10016.)

American Welding Society

B3.0 Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The truck shall consist of the following major components:

Item or System Name	See Requirement
Chassis	3.7
Tank	3.8
Pumping compartment	3.9
Pumping system	3.10
Bottom loading system	3.11.

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3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. When materials are used in the construction of the truck that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest practicable extent. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity.

3.4.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Bonding. Metal parts shall be specifically bonded together to prevent the possibility of sparking.

* 3.4.5 Metals. Unless otherwise specified herein, the tank and pumping system and all components which come into contact with the fuel shall be fabricated of nonferrous metals or 300 series stainless steel. Metals shall be free from all defects and imperfections that might affect the serviceability and operation of the truck.

* 3.4.5.1 Impregnation of castings. Aluminum castings may be impregnated in accordance with MIL-STD-276 to prevent minor weeping. Alternates to MIL-STD-276 may be used if approved by the procuring activity.

3.4.6 Elastomers. Elastomeric materials used in the tank, pumping system, bottom loading system, and all components which come into contact with the fuel shall be compatible with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.4.7 Wood. Wood shall not be used unless specifically required herein or approved by the procuring activity.

3.5 Design and construction. The truck shall be so designed and constructed that parts will not work loose in service and shall be inherently capable of

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withstanding the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The truck shall be designed and constructed for maximum ease and safety of operation.

3.5.1 Functional design. The truck shall be designed to satisfactorily receive, transport, store, and service all types of aircraft with fuels conforming to MIL-G-5572 and MIL-T-5624 under worldwide conditions. Fuels containing water shall not result in the corrosion of any part of the truck.

* 3.5.2 Reliability. The truck, including all components, shall have an MTBF (see 6.3.1, 6.3.2, and 6.6) of not less than 84 hours demonstrated at 0.9 confidence.

* 3.5.3 Maintainability. The truck shall be designed and constructed to the maximum practical extent to provide adequate clearances for repair and other maintenance and, where applicable, to permit removal and replacement of any part or assembly by removing or disconnecting only mounting bolts or fasteners, panel doors, tubing, or control cables to the part or assembly being removed.

* 3.5.4 Component accessibility. The truck shall be so designed that to the maximum practical extent components may be readily removed, serviced, and operated. Any design which renders operation unduly difficult under field operations will not be acceptable. The truck shall be designed to insure ease of production, ease and rapid installation of equipment, and ease of general maintenance and repair.

3.5.5 Handtools. The truck shall be designed and constructed in a manner that will allow general maintenance, assembly, and disassembly to be performed to the maximum practical extent with the use of common handtools.

* 3.5.6 Welders and welding

* 3.5.6.1 Welders. All welders employed in the fabrication of the truck, except the truck chassis, shall have passed the qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed. The certification that the welders have passed the qualification tests shall be on file at the contractor's facility and available for review by the procuring activity.

Standard B3.0 Standard Qualification Procedure of the American Welding Society

Welding Qualifications of the ASME

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* 3.5.6.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to test, proof, and service loadings.

3.5.7 Lubrication. Whenever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the truck design. Grease seals shall be designed and located to provide maximum accessibility for inspection and replacement and to prevent displacement, rupture, or other failure of seals due to excessive amounts of lubricant pressure. Grease seals shall be so designed that pressure due to forced lubrication will not bind rotating parts. Lubricating grooves shall be provided as required to insure uniform lubrication of internal parts. Grease fittings shall be easily accessible.

3.5.8 Air transportability. The truck shall be air transportable in accordance with MIL-A-8421 and Bulletin 518 in Types C-130, C-133, and C-141 aircraft.

3.5.8.1 Tiedowns. The truck shall be provided with sufficient tiedown points so located that the vehicle can be restrained in all directions during air transport. Tiedowns shall be designed for vehicle restraint against forces of 8g forward and backward, 2g upward, and 1.5g lateral. The attachment points of the truck shall be designed for use with attachment points in the aircraft which, in the expanse of aircraft floor, are of 10,000 pounds capacity and placed in a 20-inch grid pattern with 25,000-pound attachment points on approximate 40-inch centers along both sides of the aircraft floor. The tiedown points on the truck shall be designed and located on the basis that restraining chains will be at an angle of 30 degrees with the aircraft floor. Vehicle attachment point loading capacities shall be downgraded accordingly from the standpoint of restraining horizontal, vertical, and lateral g-loads on the vehicle.

3.5.8.2 Clearance. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure on the truck. The tiedown points shall have a minimum clear opening of 4 inches to allow attachment of a chain tiedown with a grabhook.

3.6 Performance

3.6.1 Environmental conditions. The truck shall operate satisfactorily under the following conditions without adverse effects:

a. Temperatures ranging from -65° to +125°F

b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost

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c. Fungus growth as encountered in tropical and subtropical climates

d. Salt atmosphere as encountered near bodies of salt water and in transportation on shipboard

e. Sand and dust as encountered in normal and desert conditions.

3.6.2 Pumping system operation. The pumping system shall provide for any servicing rate of 0 to 600 gpm by control of the overwing nozzles or based on the aircraft demand for single-point refueling. The pumping system shall exhibit no instability in pressure or flow rate under any operating condition. The pumping system shall function as follows:

a. Pump from the tank through the filter-separator and meters and out through the servicing hose(s) and overwing nozzle(s) at any flow rate of 0 to 300 ±10 gpm through each hose when delivering fuel through either hose individually or through both hoses simultaneously. Once the pumping system controls have been adjusted, flow shall be regulated on demand by control of the overwing nozzles.

b. Pump from the tank through the filter-separator and meters and out through the servicing hose(s) and overwing nozzle(s) at any flow rate of 0 to 60 ±10 gpm through each hose when delivering fuel through either hose individually or through both hoses simultaneously. Once the pumping system controls have been adjusted, flow shall be regulated on demand by control of the overwing nozzles.

c. Pump from the tank through the filter-separator and meters and out through the 3-inch servicing hose at any flow rate of 0 to at least 600 gpm while automatically maintaining 50 ±5 psi at the inlet to the single-point nozzle. The static pressure at the inlet to the single-point nozzle shall not exceed 60 psi. Once the pumping system controls have been adjusted, flow shall be regulated on demand downstream of the single-point nozzle. The system shall automatically recover from an externally induced no-flow condition to full flow within 15 seconds after the full-flow demand condition has been re-established.

d. During single-point operation, the pumping system shall limit the total pressure at the inlet to the single-point nozzle to 120 psi at any initial flow rate of 0 to 600 gpm when the flow is shut off in 0.5 second. The total pressure shall be measured at the inlet to the single-point nozzle with a pressure transducer and automatic electronic recording device. The locked-in pressure shall not exceed 60 psi within 15 seconds after the quick shutoff.

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e. Pump from the suction hose reel through one meter and the system tank return line to the tank at a flow rate of at least 200 gpm.

f. Evacuate 90 percent of the product in the fully flooded dispensing hoses within 1.5 minutes.

3.6.3 Pump operation. The pump shall be a self-priming centrifugal type (see 6.3.3) capable of providing the performance specified herein.

3.6.3.1 Closed discharge. The pump shall operate against a closed discharge for periods of 30 minutes without appreciable wear, damage, or overheating. The closed discharge condition shall be attained by operating the unit at maximum rated throttle setting with all manual valves closed, except the tank shutoff valve, and the system bypass operating. The fuel temperature in the pump shall not exceed 160°F during the 30-minute period with an ambient temperature and tank fuel temperature of approximately 70°F. This performance shall be attained with a maximum of 1,500 gallons of fuel in the tank.

3.6.3.2 Dry operation. The pump shall operate at maximum rated pump speed for 5 minutes without appreciable wear or damage after the unit has been completely drained.

3.6.3.3 Suction lift. The pump shall be capable of a suction lift of 15 inches Hg.

3.6.4 Tank

3.6.4.1 Static pressures. The tank shall withstand an internal static pressure of at least 3 psig without evidence of leaking.

3.6.4.2 Hydrodynamic loading. The tank shall be filled at 600 gpm through a 4-inch pipe in the tank manhole without damage.

3.6.5 Hydrostatic pressure. The piping system and all components shall withstand a hydrostatic pressure of 1-1/2 times the maximum pressure encountered during operation of the unit or a minimum of 200 psi, whichever is greater.

3.6.6 Bottom loading system. The bottom loading system shall allow the tank to be filled at any flow rate up to 750 gpm and then automatically shut off.

* 3.6.7 Mobility. When using DF-2 diesel fuel oil in accordance with VV-F-800 as the truck engine fuel, the truck shall be operable while empty and shall be capable of transporting any rated payload up to 5,000 gallons of fuel at a density of 6.6 pounds per gallon over highways at speeds of 50 mph, over unimproved roads at speeds of 40 mph, and over Belgian block pavement at speeds

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of 15 mph (see 6.3.4, 6.3.5, and 6.3.6). The truck shall be capable of the required performance with the pumping system flooded with fuel in addition to the 5,000 gallons of fuel in the cargo tank. When using grade JP-4 turbine fuel conforming to MIL-T-5624 as the engine fuel and with a payload of 5,000 gallons of fuel at a density of 6.6 pounds per gallon, the truck shall be capable of transporting the load at a speed of 35 mph on a level highway.

3.6.7.1 Suspension articulation. There shall be no interference, part failure, nor permanent distortion of any portion of the truck when it is driven over an 8-inch-high obstacle either perpendicular to the direction the vehicle is traveling or at a 45-degree angle.

3.6.7.2 Ramp clearance. The truck shall be designed to negotiate a ramp with a slope of 15 degrees in both the forward and reverse directions.

3.6.8 Truck engine starting. The truck engine shall start without external starting aids at temperatures as low as 0°F.

3.7 Chassis. The chassis shall be the manufacturer's standard, except as modified to comply with specific requirements specified herein. The chassis shall be 6-by-4 and shall conform to MIL-T-45341, type I, class E, group 2. The gross vehicle weight rating shall be at least 56,000 pounds. The required calculated gradeability as specified in MIL-T-45341 shall be 0.20. The vehicle clearance circle as defined by SAE Recommended Practice SAE J695 shall not exceed 95 feet.

* 3.7.1 Engine. The engine shall be a 2- or 4-cycle naturally aspirated diesel. Forced-air scavenged engines will be acceptable. The engine shall be capable of providing the performance requirements specified herein when operating on diesel fuel oil conforming to VV-F-800 and grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624. Any engine that is adversely affected by the difference in lubricating properties of the above fuels will not be acceptable. The engine shall be capable of providing the necessary power when using any of the above fuels, such that the normal matched progression of all transmission speeds shall occur. The engine shall not be adversely affected by long periods of idle- and low-speed operations.

3.7.1.1 Cooling system. The manufacturer's heavy-duty cooling system shall be provided.

3.7.2 Alternator. An alternator shall be furnished in lieu of a generator. The alternator shall be of sufficient capacity to maintain an adequate supply of electrical power under all operating conditions for all 12-V accessory components of the truck operating simultaneously. However, the alternator shall be not less than 60-amp capacity.

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3.7.3 Transmission. The truck shall incorporate a semiautomatic, continuous drive, automatically stepped transmission. The transmission assembly shall be of a commercial design and capacity that has been produced and supplied as a part of a commercial vehicle of comparable GVW rating. The transmission shall be one in which a portion of the total torque conversion ratios is continuously variable, but in which the remaining ratio conversions are accomplished through range control by the use of automatically stepped gear ratios that do not require more than an instantaneous power flow interruption with each selected range.

3.7.3.1 Selection of ranges of gears. The transmission shall provide for driver selection of ranges of gears required for efficient operations throughout all speeds to be encountered during operation of the truck. The shift control shall incorporate a positive indicator for each gear range selected. The transmission shall be so protected that the operator cannot inadvertently shift the truck into reverse.

3.7.3.1.1 Safety device. A safety device shall be installed to prevent starting of the engine in any selector setting other than neutral.

3.7.3.2 Torque capacity. The transmission shall have a torque capacity adequate to transmit 110 percent of the full throttle net engine torque using diesel fuel oil conforming to VV-F-800 at any engine rpm which could be encountered under any operating condition. A manually operated clutch member will not be acceptable.

3.7.4 Power takeoff. A split shaft power takeoff shall be furnished. The power takeoff shall have sufficient capacity to drive the pump for the maximum load conditions to be encountered during operation of the truck. The power takeoff and pump gearbox gear ratios shall be selected to provide the pump operations at sufficient engine speed to assure adequate engine and gearbox lubrication and engine cooling.

3.7.4.1 Power takeoff control. The power takeoff shall be air-operated and controlled by the driver while sitting in the driver's seat. The control lever shall not interfere with normal operation or access to the driver's seat by the driver entering or exiting the cab. Other controls, such as a driveshaft brake, shall be installed as required to provide a positive, smooth mechanical engagement. Any engagement device which can cause clashing of mechanical parts or heavy shock loading will not be acceptable.

3.7.4.2 Power takeoff safety devices. An interlock system shall be provided to prevent simultaneous propelling of the truck and operation of the pump. An interlock system shall be provided to prevent shifting of the power takeoff control in either direction unless the engine is at idle speed. A safety device shall be provided to prevent operation of the auxiliary throttle when the power takeoff is in the vehicle drive position.

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3.7.5 Tires and wheels. Tires shall be tire size 12.00-by-20, 14-ply with nondirectional mud and snow tread in accordance with MIL-T-12459. The tire loading capacity specified in MIL-M-8090 shall not be exceeded for individual tires. Wheels shall be of the military disc type selected from MIL-M-8090 (see 6.2). A spare tire, wheel, carrier, and tools are not required.

3.7.6 Brakes. The brakes shall conform to all Motor Carrier safety regulations for trucks without towed vehicles. Brakes shall be installed on all wheels. The service brakes shall stop the vehicle in any load condition within 30 feet from a speed of 20 mph on dry, hard, approximately level road surfaces free from loose material. The parking brakes shall hold the truck in any load condition in either direction on a slope of at least 20 degrees.

* 3.7.6.1 Airbrake emergency system. The emergency airbrake system as listed in MIL-T-45341 shall be provided.

3.7.7 Tachometer. A tachometer in the cab is not required.

3.7.8 Cigar lighter. A cigar lighter shall not be provided.

3.7.9 Horn. Either an electric or air horn shall be furnished.

3.7.10 Cab. The cab shall provide a side-by-side seating arrangement to accommodate a crew of not less than two men wearing arctic clothing. The cab shall be designed to permit ease of maintenance, easy servicing of the engine, access to the sides and top of the engine while standing on the ground, and easy engine removal. The cab shall be of the all-steel (coupe) type with a comfortably upholstered full-width seat or individual seats, the driver's seat being fully adjustable. The color of the upholstery shall be compatible with the color of the exterior finish.

3.7.10.1 The manufacturer's standard cowl ventilator or ventilators, left-hand armrest, and left-hand sunvisor shall be provided.

3.7.10.2 A rear window and domelights shall be provided in the cab.

3.7.10.3 Doors shall incorporate a vertically adjustable window panel and a pivoted ventilating window. Maximum visibility commensurate with the commercial practice for trucks shall be provided. The doors shall open wide to provide an adequate access opening for personnel entering and exiting the cab while wearing heavy arctic clothing.

3.7.10.4 The height of the chassis from the ground to the top of the cab shall not exceed 104 inches in the empty condition. The overall height of the cab shall be further reduced as required, based upon the exact configuration of the truck to be furnished and the air transportability requirements specified herein.

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3.7.11 Pintle hook. A pintle hook conforming to MS51118 shall be mounted on the rear of the truck. The pintle hook shall be mounted to permit a 25,000-pound drawbar pull without causing permanent distortion or damage to the mounting structure. The procuring activity shall specify whether or not pintle hooks are to be provided on production units (see 6.2). All prototype and first article trucks shall include installed pintle hooks, and all production trucks shall include complete provisions for the installation of pintle hooks by field activities, if not included on the production trucks.

3.7.12 Rear bumper. A full-width, heavy-duty rear bumper shall be mounted on the chassis frame. The bumper shall be constructed of steel channel section. The center portion of the bumper shall be so recessed that the pintle hook does not extend past the bumper on either side of the dished section. Adequate clearance shall be maintained for the lunette eye and a large A-frame towbar to allow sharp cornering. The outer corners of the bumper shall be rounded and shall incorporate other protection as required to minimize the hazard to personnel walking around the truck. The bumper shall be so designed that the fully loaded vehicle can be pushed without damage.

3.7.13 Wheel loading. The difference in load between the left and right wheels of any axle shall not exceed 2-1/2 percent of the load on the axle for any cargo load condition. The load on the left and right wheels of the bogie assembly may be measured in lieu of the wheels of the individual axles on the bogie assembly.

3.7.14 Bogie axle spacing. The bogie shall have an axle spacing of more than 48 inches, but not more than 54 inches.

3.7.15 Exhaust pipe and muffler. The truck shall be equipped with a spark arrestor muffler mounted underneath the forward end of the cab with the outlet at the right front corner of the truck pointing toward the curb. The muffler shall be so positioned and mounted that it cannot be easily damaged. No portion of the exhaust manifold, pipe, or muffler shall extend beyond the rear surface of the cab.

3.7.16 Lighting and wiring system. The truck lights and reflectors shall conform to and be installed in accordance with Motor Carrier safety regulations and as specified herein. Wiring, lamps, and all electrical equipment and components installed in the hazardous vapor area (see 6.3.7) shall conform to the following requirements, except the hose reel electrical components shall be in accordance with MIL-H-38209:

3.7.16.1 Wiring. Wiring for lamps shall be not smaller than 12 gage, stranded, with thermoplastic insulation with the exception of lamp pigtails which may be 16 gage, but no longer than 6 inches. Wiring shall conform to SAE standards for low-tension insulated cable.

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3.7.16.1.1 Wiring system installation. The wiring system shall be explosion-proof (vaporproof) in accordance with flammable tank vehicle industry practice. All wiring shall be enclosed in rigid, metal conduit firmly attached to the truck. All switches and circuit protection devices installed in the hazardous vapor area shall be of permanent vaporproof design or enclosed in vaporproof boxes to prevent vapor from entering the arcing device. Wiring connections to these devices shall be through the metal conduit threaded to the component or vaporproof box. Wiring junctions shall be enclosed in vaporproof boxes with threaded connections to the metal conduit. The entire system and all components shall be designed to withstand a 3-psi positive or negative pressure without leaking.

3.7.16.1.2 Protective devices. Overload circuit breakers shall be used in lieu of replaceable, fuse-type devices. All circuits shall be protected by circuit breakers. Protective devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

3.7.16.2 Lamps. All lamps installed in the hazardous vapor area shall be of permanent vaporproof and waterproof design. The lamp body shall be nonferrous and shall incorporate a screw-type fitting for making a vaporproof connection with the wiring conduit. The lens shall be heavy duty and capable of normal removal and installation without cracking or breaking. The lens shall be designed to provide a vaporproof seal with the lamp body. The lamp assembly shall be designed to withstand a 3-psi positive or negative pressure without leaking. The vaporproof seal shall not be attained by the use of cements, sealants, or similar materials.

3.7.16.2.1 Lamp bulbs. The lamp bulbs shall be of the rough-service type.

3.7.17 Batteries. The truck shall be provided with two or more Type BB-248/U (6TN), MS35000-4 storage batteries connected in parallel. The installation of, and the container for the batteries, shall provide for easy access for checking the electrolyte without removing the batteries and for easy replacement of the batteries. The batteries shall be installed forward of the back of the cab. If the batteries cannot be located forward of the back of the cab, they may be located directly behind the cab on the right-hand side.

3.7.17.1 Storage battery container. The storage battery container shall be fabricated of 300 series stainless steel and attached to the right chassis frame rail. The container shall completely enclose the batteries, prevent fuel spills which are vertically down and 45 degrees from the vertical from reaching the batteries, and permit the enclosure to breathe sufficiently to prevent buildup of gaseous hydrogen in explosive mixture ratios. The batteries shall be rigidly retained in the container. There shall be a minimum clearance

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of 1 inch between the battery terminals with connecting cables in place and the container. Provisions shall be incorporated for discharging any spilled electrolyte in the container to the ground. The battery cables shall be in conduit from the container to at least 6 inches under the cab.

3.7.18 Fenders. Fenders shall be provided over the rear wheels of the truck. The fenders shall be of heavy-gage steel and supported to withstand a static load of 400 pounds placed on top of the fenders. The outer edges of the fenders shall be rolled or reinforced with rolled-steel angle. The underside of the fenders shall be given a heavy application of high-grade vehicle undercoating. Adequate clearance shall be maintained for ease of installation of tire chains.

3.7.18.1 Skirting. Skirting may be supplied in lieu of fenders provided it is designed and constructed to provide the required protection.

3.7.19 Mudflaps. Substantial, flexible mudflaps shall be mounted behind the front wheels and the bogie, and in front of the bogie. The ground clearance of the mudflaps shall be not more than one-third of the distance from the center of the closest axle to the center of the flaps under any load condition. The mudflaps shall be mounted within 12 inches of the tires. The flaps shall be at least as wide as the tires. The flaps in front of the bogie shall be supported so as not to interfere with the running gear. If the truck is so constructed that these requirements are accomplished by means of fenders or body construction, mudflaps are not required.

* 3.7.20 Truck winterization. The truck shall incorporate a standby-type winterization kit designed to operate from an external 110-V, a-c power supply. The kit shall include at least the following components: engine coolant heater, oilpan heater, transmission heater, automatic winter front, and battery box heater. All heaters shall be immersion type and shall be thermostatically controlled to prevent overheating of the heater and the items or compartment being heated for ambient temperatures up to +35°F. The winterization kit shall be adequate for operation at -65°F.

3.7.20.1 Receptacle (110V). The 110-V electrical components used for winterizing the truck shall lead to one receptacle securely mounted and clearly labeled on the outside of the truck cab. The receptacle shall be of the male, 3-prong, twist-lock type with a spring-loaded dust cap included as part of its housing. The third wire shall be adequately grounded to the truck. A mating female plug shall be provided with each truck. The wiring shall be sized for the intended load and shall incorporate insulation impervious to water, alcohol, oil, and gasoline. All 110-V wiring shall be installed in loom. A manual reset circuit breaker shall be installed in the circuit inside the cab to afford overload protection. A red power-on indicating light shall be located on the dashpanel and clearly marked. There shall be no exposed conductor or terminals in the 110-V system which could be a hazard to personnel.

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3.7.20.2 Receptacle (12V). An external power receptacle in accordance with AN2552 shall be installed near the front right-hand corner of the cab. The receptacle shall be located in a relatively protected position and easily accessible for connection of heavy power cables without tilting the cab. The receptacle shall be wired to provide for battery charging and starting of the truck engine by an external power source. Conductors shall be sized to allow starting of the engine. A hinged cover which is held in the closed position by a spring shall be provided.

3.7.21 Compressed air system. The compressed air system shall include both a moisture ejector and an alcohol aspirator.

* 3.7.22 Driveshafts. The driveshafts, including the power takeoff, shall be protected in a manner similar to that specified for buses by Motor Carriers Safety Regulation 393.89 to prevent damage to the tank and fluid-carrying components.

3.8 Tank. The tank shall be of the single-compartment type with a 5,000-gallon liquid capacity plus at least a 3-percent expansion space. The tank shall be constructed of 5,000 series sheet aluminum of sufficient thickness to provide complete structural integrity of the tank for the life of the vehicle. The minimum sheet thickness shall be 0.187 inch for at least the bottom third of the tank. The remaining portion of the tank shall have a minimum thickness of 0.160 inch. The tank shall be elliptical or modified rectangular cross section. The tank interior shall have a bright finish. Water-stained aluminum will not be permitted.

3.8.1 Baffles. Baffles and exterior heads shall be designed, constructed, and installed in accordance with current Motor Carrier safety regulations for flammable liquid tanks. The baffles shall be designed and constructed to permit inspection and cleaning of the entire tank. Ring stiffeners or other reinforcement designs will not be acceptable in lieu of baffles.

3.8.2 Sump. A sump shall be provided at the lowest point of the tank and shall have a minimum capacity of 12 gallons to allow accumulation of solid contaminants and water. The tank shall be so designed that water will drain into the sump and not be trapped by the baffles nor stand in pockets.

3.8.2.1 Sump drain. A 2-inch drain valve shall be installed in the sump to permit complete drainage of the tank and sump. The drain valve shall be protected against flying objects and shall be easily accessible.

3.8.3 Manhole. The tank shall have a circular manhole with a minimum diameter of 20 inches. The manhole shall have a 10-inch watertight, hinged filler cover with necessary safety features to eliminate the possibility of a spark occurring

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during opening or closing of the cover. The manhole shall be equipped with a Buna N gasket, or equivalent, to insure a fuel-tight seal between the collar and the cover. The entire top of the manhole shall be removable for entrance into the tank by removing a bolt-on clamping ring.

3.8.3.1 Fusible plug. Renewable-type fusible plugs with firescreens shall be installed in the manhole cover.

3.8.4 Cleaning stubs. Three-inch cleaning stubs and caps shall be provided in the top of each tank section as separated by the baffles, except the section incorporating the manhole.

3.8.5 Tank calibration marker. A fixed marker, visible through the manhole, shall be provided inside the tank to indicate when the tank is filled with 5,000 gallons of fuel.

* 3.8.6 Auxiliary vents. The tank shall be equipped with at least two identical removable vents. Each vent shall be so designed and sized that fuel can be pumped into and out of the tank at 0 to 50 gpm with the manhole hinged filler cover, the tank loading vent, and all other tank openings closed. The vents shall not open until the pressure or vacuum in the tank reaches 0.2 psig and shall not permit the pressure in the tank to exceed 2 psig at 50 gpm flow rate, nor the vacuum to exceed 1 psig at 50 gpm flow rate. The vents shall remain closed at all other times. The vent valves shall be designed to prevent loss of cargo should the truck upset (leakage rate shall not exceed 40 milliliters per second). The vents shall be designed to avoid the possibility of malfunction or jamming. The vents shall not contain metal alloys which could corrode and cause malfunction.

3.8.7 Catwalk. A catwalk shall be provided and shall be integral with and securely attached to the tank. The catwalk shall be constructed of embossed diamond-type, expanded metal safety tread, or equivalent. The catwalk shall be at least 30 inches wide. The catwalk shall span the manhole, extend the length of the tank, and incorporate an adequate landing platform to accommodate the boarding ladder specified in 3.8.8. Means shall be provided for draining entrapped water from the catwalk. The catwalk shall be so designed that fuel will not run off of the front of the tank in the event the tank is overfilled and fuel is expelled from the manhole or vent.

3.8.7.1 Scuff guards. Tubular or formed scuff guards shall be installed at either side of the catwalk. The scuff guards shall be of sufficient height and shall incorporate structural integrity to protect the manhole and other appurtenances mounted on top of the tank in the event of overturn.

3.8.8 Boarding ladder. A tubular steel or tubular aluminum boarding ladder shall be securely attached to the rear of the tank to provide access to the

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catwalk. The ladder shall be mounted in the plane described by the rearmost surface of the bumper and in the center of the tank. The ladder shall be constructed and mounted to provide a rigid structure when scaled by a 220-pound man. A rigid step shall be suspended on the underside of the bumper to facilitate access to the ladder.

3.8.8.1 Vertical members. The vertical members of the ladder shall be 18 to 20 inches apart. The vertical members shall extend at least 1 foot above the top surface of the tank and 18 inches horizontally forward of the ladder and then be firmly attached to the tank structure.

3.8.8.2 Safety features. The ladder shall be so designed that it does not interfere with access to the pintle hook and so that the pintle hook will not be a safety hazard to personnel using the ladder. The ladder assembly shall be inherently safe and convenient for operating personnel.

3.8.8.3 The portion of the ladder which extends beyond the top portion of the tank shall be removable by one man with common handtools within 5 minutes for preparing the truck for air transport.

3.9 Pumping compartment. The pumping system and all its components, with the possible exception of the pump and suction stub, shall be housed in a completely enclosed compartment located between the cab and cargo tank. All pumping system controls, dispensing hoses, meters, and the suction stub shall be accessible for normal operation from the left or driver's side of the truck.

3.9.1 Pumping compartment construction. The pumping compartment shall be of single construction and designed for heating all pumping system components using an external hot air heater. All joints shall be tight. The compartment shall be constructed of sheet steel with a minimum thickness of 16 gage (0.062 inch). All panels shall be integrally constructed with reinforcement and permanently welded. At least the roof and side panels shall be removable. All doors and hinged sections shall have full-length, piano-type hinges with corrosion-resistant hinge pins. The compartment assembly shall incorporate necessary structural integrity to prevent bending, warpage, or other type damage due to vibration and other forces encountered during normal operation. Sharp edges and corners which could be a hazard to operating personnel will not be acceptable.

3.9.1.1 Roof. The pumping compartment roof shall be hinged in the center to provide access to and removal of components on either side of the compartment. The roof shall also be removable as an assembly. The roof shall be held in position by hand-operated latching devices that allow access to either side of the compartment or removal of the complete roof by three men without tools in

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3 minutes. The roof shall support three men standing on top of the compartment without permanent distortion or other damage. The roof panel shall be provided with lifting handles to allow opening of the hinged half-sections and removal of the complete panel.

* 3.9.1.2 Doors. Doors shall be provided on the left or driver's side of the compartment. The doors shall be mounted on vertical hinges and shall provide convenient access to the servicing hoses, control panel, and control valves. The doors shall be fastened to the hinges by bolts, nuts, and lockwashers or bolts and self-locking-type nuts. The doors shall provide an opening substantially the full length of the compartment and at least 36 inches high. At least one door, one-half the length of the compartment and at least 30 inches high, shall be provided on the right side of the compartment. This door shall provide access for inspection and minor maintenance of the filter-separator sump manual and automatic controls, the pumping system automatic control valves, inspection of the pumping system and its components for leaks, and shall allow servicing of components as required. Both doors on the right and left sides shall be so located that a man can enter the compartment through the door openings without difficulty. Automatic engaging devices shall be provided to hold the doors in the open position. The doors and hold-open devices shall be constructed to prevent damage under high winds.

* 3.9.1.2.1 Door latches. Door latches shall be rigidly constructed and mounted to prevent bending, distortion, and other type failures during long periods of operation. The latches shall be held in place by bolts, nuts, and lockwashers or bolts and self-locking-type nuts. The latches shall be of the quarter-turn release type with a positive-locking means that will prevent the latches from inadvertently releasing due to vibration or other forces. Two locking pins per latch shall be provided at the top and bottom of the door. Locking pins shall engage the side panel structure rather than the opposite door. The latches shall be designed for operation by personnel wearing arctic mittens.

3.9.1.3 Floor. The floor shall fully enclose the bottom of the pumping compartment. The floor shall be constructed of sheet steel with a minimum thickness of 12 gage (0.109 inch). The floor panel shall be reinforced to insure necessary structural integrity. The floor shall be sloped and shall incorporate sufficient drain holes to insure that spilled fuel will not accumulate in the compartment.

3.9.1.4 Heater adapters. One 6-inch air duct coupling conforming to Drawing 50C24046 shall be mounted on each side of the pumping compartment, except the couplings may be fabricated of mild steel. The couplings shall be flush-mounted and provided with a cover or louvers. If a cover is used, a chain or other means of attachment shall be provided to prevent loss. The couplings shall be mounted as low as possible to provide best natural circulation around the hose reels and other components. The coupling on the right side shall be located to

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direct the incoming hot air onto the filter-separator sump and automatic sump drain valve. The couplings shall not be mounted on the front of the compartment. The mounting positions shall allow adequate clearance for ease of attachment of the heating ducts.

3.9.1.5 Ventilating provisions. The compartment shall be designed to provide for natural air circulation and purging to the maximum practical extent under all conditions, except extreme low temperature operation. There shall be no pockets which will allow the accumulation of heavy fuel vapors. There shall be no ventilating openings in the front panel of the pumping compartment (facing the cab).

3.9.1.5.1 Side panel top louvers. Each side panel shall have at least one louvered ventilating section located near the top of the compartment. The louvered sections shall extend nearly to the ends of the compartment to provide adequate distribution of fresh air entering the compartment interior. The louvered section(s) in the top of each side panel shall provide at least 2 square feet of open area. Panels shall be provided for covering the louvered sections on the inside of the compartment for extreme low temperature operation and compartment heating. The panels shall be hinged or shall have storage provisions. The panels shall be firmly attached to the pumping compartment structure in both the stored and in-use positions.

3.9.1.5.2 Bottom ventilation openings. Ventilation openings shall be provided in the floor of the compartment. The openings shall be located toward the outside of the compartment and shall prevent the accumulation of hazardous vapors in corners or other parts of the compartment. Each side of the floor area shall have at least 2 square feet of open ventilation area. Panels shall be provided for covering the ventilation openings for extreme low temperature operation and compartment heating. The panels shall be hinged or shall have storage provisions. The panels shall be firmly attached to the pumping compartment structure in both the stored and in-use positions. The openings shall be so designed that excessive dirt will not be thrown into the compartment from the wheels since the panels will be normally open. The nozzle storage compartment shall protect the nozzles from road dirt as thrown by the wheels.

3.10 Pumping system. The pumping system shall include the tank-to-pump suction line, pump, filter-separator, meters, dispensing hose reels, dispensing hoses, nozzles, hose evacuation system, system bypass line, control system, control panel, and defuel kit. Operation of the pumping system for fueling and defueling aircraft shall not involve the use of external sensing hoses or other devices on any other equipment other than that specifically referred to herein. There shall be no electrical equipment or devices associated with or incorporated as a part of the pumping system, except as specifically referenced herein. The pumping system shall operate for continuous periods of several hours without degrading effects.

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3.10.1 Tank-to-pump suction line. The tank-to-pump suction line shall be designed to provide adequate flow and smooth operation of the pump under all conditions specified herein. The suction line shall incorporate a line strainer and tank shutoff valve. A flexible coupling may be provided if required by the design of the tank and pumping compartment mountings. The suction line shall terminate in the bottom of the tank, not the sump, so that contaminants accumulated in the sump will not enter the pump.

3.10.1.1 Line strainer. The suction line shall incorporate a Y-type line strainer. The 4-mesh screen shall be rigidly constructed and adequately sized to allow a 600-gpm flow at minimum pressure drop. The screen shall be easily removable for cleaning and inspection.

3.10.1.2 Tank shutoff valve. The valve shall be of the quarter-turn to full-on to full-off type. The valve shall be located between the tank and flexible coupling or between the tank and suction stub if the flexible coupling is not furnished. The valve control shall be easily accessible from the left or driver's side of the truck without requiring the operator to crawl under the truck.

3.10.1.3 Flexible coupling. If required to provide relative motion between the pumping system and the tank, a flexible coupling shall be provided in the tank-to-pump suction line. The flexible coupling shall be located downstream of the tank shutoff valve. The flexible coupling shall withstand an internal pressure up to three times the working pressure or 50 psi whichever is greater, and a vacuum of 0 to 15 inches Hg without leakage or other type of failure.

3.10.1.4 Pump shutoff valve. The pump shutoff valve shall be of the quarter-turn to full-on, to full-off type. The valve shall be located downstream of the pump and before the bypass line or any other branch lines.

3.10.2 Pump. The pump shall be of the self-priming, centrifugal, aircraft fuel dispensing type driven directly by the power takeoff (see 6.3.3). The pump and pumping system shall provide the performance specified herein when handling fuels conforming to MIL-G-5572 and MIL-T-5624.

3.10.2.1 Pump seal failure. The pump shall be so constructed and positioned that fuel will not enter into any gearbox or transmission in the event of pump seal failure.

3.10.2.2 Pump drain. A drain plug or valve shall be provided for completely draining the pump.

* 3.10.3 Suction stub. The tank-to-pump suction line shall incorporate a suction stub upstream of the Y-strainer. The suction stub shall be a nominal 3-inch size with a manually operated shutoff valve. The stub shall be provided

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with a 3-inch, class I, female, cam-locking quick-disconnect coupling half with a dust plug and a gasket in accordance with MIL-C-27487. The dust plug shall have a security cable in accordance with MIL-W-1511, except with a clear nylon jacket. The suction stub and shutoff valve shall be mounted outside of the pumping compartment and shall be easily accessible to an operator standing on the ground at the left or driver's side of the truck. Adequate accessibility shall be provided for connection of a suction hose or the defuel kit.

3.10.4 Filter-separator. A filter-separator with vertically mounted elements having a rated capacity of 600 gpm shall be provided. The filter-separator shall meet the performance requirements of MIL-F-8901; except if double-stacked elements are provided, then the allowable differential pressure shall be 15 psi in lieu of 10 psi. Thirty military standard elements in accordance with MIL-F-52308 shall be provided. Each element or stack of two elements shall be encased by a canister assembly which includes the water separation stage, a 100-by-100 mesh 0.0046-inch diameter monel wire screen coated on both sides with polytetrafluoroethylene molding plastic in accordance with L-P-403, and a minimum of 14 gage (B & S) aluminum alloy perforated sheet which provides protection for the coated monel screen and structural strength in conjunction with the canister retention provisions for holding the elements in place with a liquidtight seal. The canister shall be capable of retaining 0.5 inch of water. If the elements are double stacked, an intermediate seal plate in accordance with Drawing B13201E6470 shall be provided. Provisions shall be included which assure that the canister is centered about the element or elements. The filter-separator shall be so installed in the system that all fuel will pass through the filter-separator in only one direction and that being the direction to provide clean, dry fuel.

3.10.4.1 Pressure vessel. The pressure vessel shall be designed and fabricated in accordance with the ASME Boiler and Pressure Vessel Code. The pressure vessel shall be designed for a working pressure of the maximum pressure encountered during operation of the unit. The pressure vessel shall be hydrostatically tested at 1-1/2 times the working pressure, but not less than 200 psi. The pressure vessel flange gasket shall be of the O-ring type. The main vessel and the cover shall be provided with lifting handles capable of lifting the entire filter-separator assembly.

3.10.4.2 Air eliminator. An air eliminator shall be provided to allow automatic purging of air from the pumping system. The air eliminator shall be vented into the cargo tank.

* 3.10.4.3 Water slug control. A water slug control shall be provided which will stop the fuel flow when a predetermined water level is reached in the vessel and then permit fuel flow when the water level is lowered to a predetermined level. The effluent fuel quality shall be in accordance with MIL-F-8901 when influent fuel contains up to 5 percent undissolved water.

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3.10.4.3.1 Automatic water drain. The water slug control shall contain an automatic water drain to automatically discharge water as it accumulates without interfering with the normal function of the pumping system. The drain shall extend outside of the pumping compartment.

* 3.10.4.3.2 Float-actuated selector valve. The float-actuated selector valve, used in conjunction with the water slug control and automatic water drain shall include provisions which permit manual movement of the float through its complete range when installed and the system is pressurized.

* 3.10.4.4 Manual drain. In addition to the automatic water drain, the sump of the filter-separator shall be equipped with a 3/4- to 1-inch manually operable valve which must be held to remain in the open position. The drain shall extend outside of the pumping compartment.

3.10.4.5 Differential pressure gage. The filter-separator shall be equipped with a differential pressure gage having a diameter of 4-1/2 inches graduated in 1-psi increments throughout the range of 0 to 100 psi. The gage shall be connected to continuously monitor pressure across the complete filtration media including the water arresting stage. The gage shall have an accuracy of within 1 percent of the full scale range over the full scale range. The gage shall be protected against pressure surges.

3.10.4.6 Canister retention. Canister retention shall be provided which assures that the element seals are maintained under all operating conditions, elements are not damaged during transit, elements are easy to remove and replace, and canisters are easy to remove and replace.

3.10.4.7 Element seals. The surfaces that fit the element seals shall have an outside diameter of 1.184 $\pm$0.002 inch with the outer corner rounded to a 1/8-inch radius.

3.10.5 Meters. Two meters conforming to MIL-M-6404 shall be provided. The meters shall withstand the pumping system hydrostatic pressures without leakage or damage. One meter shall be installed at each dispensing hose reel inlet on the left (driver's side) of the truck. The meters shall be clearly visible and within easy reach of an operator standing on the ground at the control panel. The meters shall be so installed that only the fuel passing through the meters is actually dispensed to the aircraft. The meters shall be so installed that the fuel passing through each hose reel, individually or simultaneously, is metered individually. The meters shall function in combination to meter the flow through the 3-inch dispensing hose for high-flow single-point servicing. The meters shall be so installed that all fuel entering the pumping system during defuel is metered prior to delivery to the cargo tank.

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3.10.5.1 Flow limiting. The pumping system shall limit the maximum flow through each meter to 350 gpm under all operating conditions including maximum throttle, single-point servicing with a nozzle pressure of 10 to 15 psi.

3.10.6 Dispensing hose reels. Two dispensing hose reels of the single-wrap design shall be provided in the pumping system. The hose reels shall conform to MIL-H-38209, except as specified herein. One hose reel shall be provided with 60 feet of 2-inch ID collapsible hose conforming to MIL-H-26521 for servicing fuel at flow rates of 0 to 300 gpm. One hose reel shall be provided with 60 feet of 3-inch ID collapsible hose conforming to MIL-H-26521 for servicing fuel at flow rates of 0 to 600 gpm. Hose reel inlet connections shall be of the grooved type. When specified (see 6.2), 1-1/2-inch ID hose by 60 feet in length in accordance with MIL-H-6615 shall be provided on production units in lieu of the 2-inch hose in accordance with MIL-H-26521. When 1-1/2-inch hose is provided, a 2-inch by 1-1/2-inch reducer shall be provided for the hose reel connection. All applicable flow tests shall be conducted using the 2-inch ID hose and the Type MD-2 nozzle.

3.10.6.1 Overwing nozzles. One Type MD-2 nozzle conforming to MIL-N-26978 shall be provided for the 2-inch ID hose. If the 1-1/2-inch ID hose (see 6.2) is required instead of the 2-inch ID hose, then one Type MD-1 nozzle conforming to MIL-N-26978 shall be provided. Unless otherwise specified (see 6.2), a Type MD-2 nozzle conforming to MIL-N-26978 with a quick-disconnect coupling capable of mating with the inlet half of the single-point nozzle coupling strainer assembly conforming to MIL-C-38479 shall not be provided for the 3-inch hose.

3.10.6.2 Single-point nozzle. A single-point nozzle conforming to MIL-N-5877 shall be provided for the 3-inch hose.

3.10.6.3 Single-point coupling assembly. A GRK-15/S32R coupling conforming to MIL-C-38479 shall be provided for connecting the single-point nozzle to the 3-inch hose.

3.10.6.4 Shutoff valves. A shutoff valve shall be located upstream of each hose reel. The shutoff valves shall be hydraulically or pneumatically controlled by individual controls located on the pumping system control panel. The valves shall open when the control is placed in the ON position and the hoses are unreeled and shall remain closed (pressure tight) at all other times including when the pump is not operating.

3.10.6.5 Pressurization safety controls. Each hose reel shall be equipped with a device to prevent pressurization of the hose reel, unless the hose is completely unwound. The device shall consist of an idler gear and drivescrew

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driven by the hose reel drum gear. The drivescrew shall actuate a lever which mechanically actuates a pilot valve as the hose becomes completely unwound. The pilot valve shall be in series with the shutoff valve control pilot on the control panel. The safety control shall prevent damage to the hose reel due to pressurization under any condition. The safety control shall be installed as a unit and an accessory to the hose reel. The mounting shall be adjustable to insure proper mating of the gears. The safety control shall be adjustable to allow proper actuation of the pilot for any length of hose installed on the hose reel from 35 to 75 feet. The safety control assemblies shall be designed and constructed to provide long life without excessive maintenance and shall be inherently fail-safe.

3.10.7 Hose evacuation system. The evacuation system shall evacuate 90 percent of the fuel in the fully flooded discharge hoses within 1.5 minutes and prevent fuel from flowing back into the hoses. Hose evacuation shall be accomplished by placing the control panel selector on evacuate. No other manual operation will be required, except adjustment of the engine throttle.

3.10.8 Bypass line. The bypass line shall relieve the pump under low-flow and no-flow conditions. An automatic bypass valve which integrates with the control system shall be provided to achieve the performance required herein. The bypass line shall be located upstream of the filter-separator and shall terminate in the cargo tank. The bypass shall prevent engine speed changes of more than 350 rpm between full-flow and no-flow for any pumping mode, except defuel. All bypass pilot valves shall be inactivated while defueling with the exception of the maximum pressure pilot specified in 3.10.10.5 in order to prevent any possibility of bypassing fuel.

3.10.9 System tank return line. In addition to the system bypass, a tank return line shall be provided. The return line shall allow return of fuel to the tank during defuel and evacuate and shall not be in use any other time. The return line shall be so designed that all fuel evacuated from the hose will pass through the meters in the reverse flow direction, and all fuel entering the suction stub will pass through one meter in the normal flow direction.

3.10.9.1 Flexible couplings. If required to provide relative motion between the pumping system and the tank, flexible couplings shall be provided in the system bypass line and the system tank return line. The flexible couplings shall withstand an internal pressure of up to three times the working pressure or 200 psi, whichever is greater, without leakage or other type failure. All pipe openings into the pumping compartment shall be protected to limit air passage around the pipes into or out of the compartment.

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3.10.9.2 Check valves. The system bypass line and the system tank return line as well as any other piping or tubing between the tank and pumping system, except the tank-to-pump suction line, shall incorporate check valves to prevent backflow from the tank when maintenance is performed on the pumping system. The check valves shall be located between the tank and the flexible coupling or the first detachable point.

3.10.10 Control system. The control system shall be hydraulically operated. The system shall automatically provide the pressure control and flow relief functions to accomplish the performance requirements specified herein. The control system shall prevent pump speed surges, pressure surges, and any other type of pumping system instability. The system shall automatically regulate to the required flow conditions as determined by the overwing nozzles and the aircraft fuel system downstream of the single-point nozzle. The control system shall include the system bypass valve, pressure sensing venturi, hose reel shutoff valves, filter-separator water discharge slug control, crossover valve, evacuation and defuel control valves, and the associated automatic pilot valves and operating controls. The pressure-sensing venturi shall be located downstream of the filter-separator. The hose reel shutoff valves may be pneumatically operated and a part of the pneumatic power system. The hydraulic control system shall relieve to the pump suction, not to atmosphere.

3.10.10.1 Selector valve. The selector valve shall be the primary operating control of the pumping system. The selector valve shall have five positions: low flow, high flow overwing, high flow single point, defuel, and evacuate. The selector valve shall provide for automatic adjustment of the pumping system controls to accomplish the corresponding pumping mode. A stop shall be provided such that the selector valve cannot be rotated through the defuel position when going from evacuate to the pumping positions.

3.10.10.2 Operating controls. In addition to the selector valve, the operating controls shall include only the hose reel shutoff controls, suction stub valve, and the tank shutoff valve. The suction stub and tank shutoff valves shall be the only manually operated valves in the system and shall require operation only for defuel.

3.10.10.3 Crossover valve. The automatic crossover valve shall be located downstream of the meters. The crossover valve shall allow flow from both meters to be serviced through the 3-inch hose reel for single-point servicing at 0 to 600 gpm. The crossover valve shall remain closed at all other times.

3.10.10.4 Control system maintenance adjustment. The control system shall be so designed that all control adjustments can be accomplished in a specified sequence, and all adjustments are independent. The control system shall be so designed that all adjustment points are accessible without the removal of any part or components.

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3.10.10.5 Safety provisions. The pumping system shall be so designed that all valves and controls including the throttle can be adjusted in any position without causing damage to any portion of the truck under any operating condition, or exceeding 60 psi at the single-point nozzle during single-point servicing or 350 gpm through each meter. This requirement shall be complied with when the system is properly adjusted and when all adjustments are purposely positioned to the most unfavorable conditions. A nonadjustable pilot shall be provided which will not allow the normal maximum pumping system operating pressure to be exceeded by more than 10 psi under any condition.

3.10.10.6 Single-point nozzle pressure gage. The pressure gage shall continuously monitor pressure at the venturi throat or other point in the system which most closely simulates the actual single-point nozzle pressure. The gage shall be utilized to monitor single-point nozzle pressure. Calibration tests shall be conducted at flow rates from zero to the maximum single-point flow rate at an externally induced nozzle pressure of 50 $\pm$5 psi and at the maximum flow with the nozzle pressure relieved to the attainable minimum. There shall be no more than 5 psi difference between the single-point nozzle pressure gage at the venturi throat or other point in the pumping system and a test gage used to measure pressure at the inlet to the single-point nozzle for any flow or the static condition. The pressure gage shall be mounted on the control panel and marked SINGLE-POINT NOZZLE PRESSURE. The vehicle technical order shall include a table listing the single-point nozzle pressure measured by the gage on the control panel and the actual single-point nozzle pressure measured at the inlet to the single-point nozzle for flow rates of 0 to 700 gpm in 50-gpm increments. The gage shall be 4-1/2 inches in diameter with a 0-to-100 psi scale with 1-psi graduations. The gage shall have a 1-percent accuracy of the full-scale range over the full-scale range. The gage line shall incorporate provisions to insure the gage is not subjected to negative pressures.

3.10.11 Control panel. The pumping system shall incorporate a control panel on the left or driver's side of the pumping compartment. The control panel shall be located in the compartment. It shall be within easy reach of and clearly visible to an operator standing on the ground. The control panel shall be easily removable or easily accessible from the rear for maintenance purposes. The control panel shall contain the auxiliary throttle control, tachometer and combination hourmeter, selector valve control, hose reel shutoff valve controls, venturi throat pressure gage, filter-separator differential pressure gage, pump discharge pressure gage, hose reel control switches, pumping compartment light switch, and the engine emergency switch. The controls shall be arranged for maximum convenience to the operator. The hose reel controls shall be located left and right to correspond to the hose reels.

3.10.11.1 Auxiliary throttle. The auxiliary throttle shall provide for control of engine speed during all pumping system operations. The throttle

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shall allow the operator to control the engine at any desired speed, from idle through maximum rated pump rpm as required to conform to the performance requirements specified herein. The throttle shall provide for sufficient movement and smooth operation to allow the operator to select a previously determined operating speed within ±50 rpm with no difficulty. A turn-to-lock mechanism shall be provided to allow the throttle to be locked at any desired setting.

3.10.11.2 Tachometer. The control panel shall incorporate a tachometer and combination hourmeter to continuously read engine speed. The maximum allowable engine speed during operation of the pumping system shall be indicated by a red line on the face of the tachometer.

3.10.11.3 Pump discharge pressure gage. The pump discharge pressure gage shall continuously monitor pressure at the outlet to the pump. The gage shall be 4-1/2 inches in diameter with a 0 to 150 psi scale with 1-psi graduations. The gage shall have a 1-percent accuracy of the full-scale range over the full-scale range. All three pressure gages shall incorporate restriction plugs as required to dampen oscillations and provide stability of the indicating needle.

3.10.11.4 Engine emergency switch. The control panel shall incorporate an engine emergency manual cutoff switch to provide for immediate and complete shutdown of the truck engine from any operating condition without the operation of any other controls. The switch shall be marked by a red circle 1 to 2 inches in diameter or otherwise conspicuously marked.

3.10.12 Defuel kit. The defuel kit shall consist of a hand-driven hose reel and enclosure complete with hose, single-point nozzle, and coupling assembly which can be mounted on the left (driver's side) of the vehicle and attached to the suction stub. The quantity of defuel kits to be provided and specific units requiring installation shall be as specified by the procuring activity (see 6.2). All prototype and first article trucks shall include installed defuel kits.

3.10.12.1 Defuel kit installation. The defuel kit shall be designed for installation behind the lower left rear corner of the pumping compartment. The kit shall be attached to the chassis frame rail or other supporting structure. The kit shall be designed for field installation by two men in 1/2 hour using handtools only, and without drilling or welding. The defuel kit shall be complete to the extent that no additional hardware is required for installation or operation. The defuel kit shall include the proper configuration length of pipe for connection to the suction stub. The suction stub cam lock adapter may be removed for connection of the pipe, however, the adapter will be furnished with the truck even when the defuel kit is installed. The defuel kit

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shall not interfere with or be damaged by normal operation of the truck. The defuel kit shall not extend beyond the side of the vehicle. The defuel kit shall be considered a component of the truck.

3.10.12.2 Hose reel. A multiwrap hose reel conforming to the design requirements of MIL-H-38209 shall be provided as part of the defuel kit. The reel shall be powered by a handcrank in lieu of an electric motor. The hose reel shall contain 50 feet of 2-1/2-inch ID noncollapsible hose conforming to MIL-H-4441. The hose shall withstand a defuel flow rate of at least 200 gpm. The hose reel inlet connection shall be of the grooved type. It is not necessary to conduct the preproduction tests of MIL-H-38209 on the defuel kit hose reel.

3.10.12.3 Single-point nozzle. In addition to the nozzle specified in 3.10.6.2, a single-point nozzle conforming to MIL-N-5877 shall be provided for the 2-1/2-inch suction hose. The single-point nozzle shall incorporate a suction breaker capable of restricting vacuum to 5 psig at the nozzle.

3.10.12.4 Single-point coupling assembly. In addition to the coupling assembly specified in 3.10.6.3, a GRK-15/S32R coupling conforming to MIL-C-38479 shall be provided for connecting the single-point nozzle to the 2-1/2-inch suction hose.

3.10.12.5 Handcrank. A handcrank shall be provided on the suction hose reel for rewinding the hose. The handcrank shall be removable and used only during rewinding of the hose. A storage position for the handcrank shall be provided. The hose reel shall be so designed that the hose can be rewound by one operator. The hose reel shall be of the bottom wind type or shall incorporate other design features so the handcrank rotation does not interfere with the hose as it is being rewound. The hose reel shall be designed for ease of operation and convenience to the operator.

3.10.12.6 Housing. The defuel kit shall include a housing to fully enclose the hose reel. The housing, including the door, shall generally conform to the requirements of 3.9. The defuel kit and its components shall be protected from the road dirt and objects thrown by the wheels. The housing shall incorporate provisions for storing the nozzle.

3.10.13 Piping. The piping layout shall be designed to facilitate maintenance on the pumping system and removal of components with minimum disassembly of piping. Grooved-type couplings shall be used to the maximum practical extent with due regard to low-temperature operation. Piping sections shall be easily removable for maintenance and repair. To the maximum practical extent, piping connections to major and minor components shall be of the grooved type. The

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piping shall be designed to facilitate draining to the maximum practical extent. The large pipe sections and pipe sections which could be subjected to excessive movement during operation shall be firmly mounted or braced. Piping wall thickness shall be a minimum of schedule 40 in both the pumping system and the bottom loading system.

3.10.13.1 Tubing and conduit. Wherever possible, static pressure and gage-sensing tube lines shall be installed in groups and mounted on blocks which provide for separation of the lines and mounting of multiple lines in one place on any flat surface.

* 3.10.14 Static cable reels. Two static discharge grounding reels shall be provided. The reels shall conform to MIL-R-83232, except that instead of a W-C-440 type PCL, item No. 1 electrical clip, one reel shall be equipped with an MS27610 connector, and the other reel shall be equipped with a plug conforming to Drawing 69B52267. The reels shall be installed side-by-side with the base in a horizontal position in a readily accessible location to the rear of the pumping compartment on the left-hand side of the truck. The location of the reels shall permit easy withdrawal and recovery of the cables. The reels shall be bolted to the truck, and the resistance between the reel and the chassis frame and tank shall not exceed 0.5 ohm.

* 3.10.15 Sampling device. A sampling device shall be provided in the outlet piping to permit sampling the fuel downstream of the filter-separator. If possible, the sampling device should be located in a vertical pipe. The sampling device nipple shall extend at least 1/4-inch beyond the inside wall of the pipe from which the sample is obtained. Adequate space shall be provided to permit connection of the Millipore Corporation, Bedford, Massachusetts 01730 fluid sampling kit, catalog No. XX 64 037 00 for solids and water samples. The sampling device shall consist of the following:

* a. Nipple, 1/4-inch nominal, schedule 40, aluminum pipe 2 inches long with one end threaded with 1/4-18 NPT.

* b. Valve, ball-type; 1/4-inch; internal 1/4-18 NPT both ends; aluminum-alloy body; highly polished, stainless-steel ball; 300 series stainless-steel stem; teflon seals; cadmium-plated malleable iron handle; cadmium-plated carbon steel handle nut; cadmium-plated carbon steel handle stops; quarter turn of handle and valve goes from full open to full close; handle stops limit rotation to 90 $\pm$5 degrees; pressure loss shall not exceed 1.1 psig at 1.1 gpm of water.

* c. Quick-disconnect, Snap-Tite, Inc., part No. APEC4-4M, manufactured by Snap-Tite, Incorporated, Titusville Road, Union City, Pennsylvania 16438 or a coupling of equal construction and characteristics, which shall connect to the sampling kit quick-disconnect.

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* d. Dust plug, Snap-Tite, Incorporated, part No. AMPE-4, manufactured by Snap-Tite, Incorporated, or equal, to fit the APEC4-4M quick-disconnect with a 1/16-inch-diameter security wire rope in accordance with MIL-W-1511, except with a clear nylon jacket.

* e. Teflon tape shall be used on the threads that connect the nipple, valve, and quick-disconnect together.

3.11 Bottom loading system. The truck shall be equipped with a fully automatic bottom loading and vent system. The major components of the system shall be an inlet manifold and control valve, float control shutoff, and an air-operated vent. The system shall be designed for a maximum amount of operating flexibility, ease of operation and maintenance, and adjustment with common handtools. The system, including the automatic vent, shall be designed fail-safe and to prevent damage to the truck or any of its components if the bottom loading system malfunctions or is operated incorrectly with the exception of the level control manual override. All components of the system shall be removable without entering the tank. Air supply, control lines, and fuel lines shall be routed external to the tank and neatly mounted and grouped. The lines shall be protected from normal hazards associated with operation and maintenance of the truck. There shall be no electrical equipment or operation as a part of or associated with the bottom loading system. The control panel, manifold, and all other components shall be protected from road dirt and objects thrown by the wheels, although they need not be fully enclosed. The control panel and manifold shall be located behind the left rear fender of the truck.

3.11.1 Pneumatic power system. The bottom loading system shall be pneumatically operated from the truck airbrake system. Safety provisions shall be incorporated that will prevent loss of air in the truck airbrake system in the event of failure of any portion of the bottom loading system. Components which operate with fuel and truck air under pressure shall be so designed that leakage of fuel into the air system or vice versa cannot occur under any operating or malfunction condition including seal or part failure. The pneumatic power system shall be essentially a static system which releases air only when controls are operated and only to the extent necessary to purge components. Under no condition shall air be vented into the tank. The bottom loading system, both pneumatic and fuel components, shall be protected with screens, filters, or other devices to prevent contaminants from entering critical components. All pneumatic components shall withstand a pressure of four times the maximum pressure encountered during operation of the truck.

3.11.2 Inlet manifold. The inlet manifold shall extend from the inlet receptacle to the bottom of the tank. The manifold shall terminate in the bottom of the tank in a compartment separated by the baffles from the inlet to the tank-to-pump suction line. The manifold shall be a nominal 4-inch size

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and shall allow tank filling rates up to 750 gpm, gravity discharge rates up to 100 gpm, and a pump discharge off-loading rate of 450 gpm. The manifold and all its components shall be designed for static pressures up to 150 psi. The pressure drop through the manifold shall be kept to a minimum.

3.11.2.1 Inlet receptacle. The receptacle shall be located on the left or driver's side of the truck in a convenient location to allow attachment of the single-point nozzle and hose. The receptacle shall mate with a Type D-1 single-point nozzle conforming to MIL-N-5877 and shall incorporate a high-strength MS24484-2 adapter and a dust cover conforming to MIL-A-25896. The inlet receptacle shall be designed to open by attachment and operation of a Type D-1 nozzle. The receptacle and associated piping shall be substantially mounted to avoid failure due to the nozzle, fuel-filled hose, and loads caused by rapid increases and decreases in pressure.

3.11.2.2 Manual valve. A 4-inch manual valve shall be provided. If the automatic control valve is located in the manifold rather than in the tank, the manual valve shall be located between the tank and automatic control valve. The manual valve shall be of a type that can be adjusted to at least eight partially open positions for checkout of the pumping system pressure control system. The manual valve shall be adjusted to the full-open position for all operations of the bottom loading system.

3.11.2.3 Control valve. A 4-inch control valve shall control the flow into and out of the tank. The valve shall be remotely operated. The valve shall be controlled open or shut when the tank is not full by manually operating a pneumatic control. An air cylinder shall be provided to actuate the lever controlled valve. The control valve shall be of the balanced type and spring loaded to close when air pressure is relieved from the cylinder. The valve shall be so designed that the air cylinder will have sufficient force to open the valve under all conditions and incorporate adequate structural integrity that full air pressure will not cause failure of any part in the event the valve is held shut. The valve and control mechanism shall be so designed that the valve will not close in less than 3 seconds when the air supply is shut off by either the manual or lever control. The valve shall shut off in not more than 6 seconds when the manual valve is turned off and not more than 20 seconds at 600 gpm when the float level actuates. A metal box or other type structure shall be mounted in the tank over the manifold outlet or the tank type valve, if provided, in order to prevent the fuel from spraying unrestricted into the tank.

3.11.3 Level control. The level control shall automatically shut off the manifold control valve when the tank is filled with 5,000 gallons of fuel. The float-operated level control shall repeatedly shut off the fuel flow within $\pm$1 inch of the preset level at any flow rate of 100 to 750 gpm. The

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level control shall be adjustable to allow a preset level of +0 -6 inches of the 5,000-gallon level. The level control shall shut off the air supply to the manifold control valve air cylinder and vent the entrapped air downstream of the float control to allow the manifold control valve to close. The float shall be nonsinkable and designed to provide reliable operation under all conditions. A flow sensing-flow interruption type level control may be utilized provided it can perform the same functions and incorporate at least equivalent performance in the bottom loading system.

3.11.3.1 Level control tester. A level control tester shall be provided to determine proper functioning of the float control. The tester shall be designed for operation during the tank filling operation. The tester shall function by allowing fuel to flow from the inlet manifold into the float chamber, thus actuating the level control and manifold control valve to simulate a full tank shutoff. The tester control valve shall be located in the flow line running from the manifold to the float chamber. The valve shall be of the holddown-to-operate or deadman type to shut off flow to the chamber, unless held in the open position. Once the test operation has been completed and the holddown valve released, the float chamber shall automatically drain and thus reopen the manifold control valve to allow the filling operation to be completed. The complete cycle shall require not more than 1 minute with manifold flow rates as low as 100 gpm. The flow line shall be located on the manifold such that the manual valve (3.11.2.2) may be closed when the level control function is tested.

3.11.3.2 Level control manual override. The bottom loading system shall incorporate necessary provisions for overriding the level control. The override shall be used to supply air to the manifold control valve. The override shall be used to open the manifold control valve to allow gravity or pump discharge of the tank contents through the manifold when the tank is initially completely filled. Once a sufficient quantity of fuel has been discharged to allow the level control to provide air pressure to the manifold control valve, actuation of the override shall no longer be required. The override control valve shall be of the holddown-to-operate or deadman type to shut off air pressure to the air cylinder unless held in the open position. The override control shall provide shutoff of the air supply and vent the entrapped air downstream of the override control to allow the manifold control valve to close when the holddown control is released. To avoid use of a two-chamber air cylinder for the manifold control valve, the air supply from the override control shall be connected to the float control valve vent port. The float control venting will then be back through to the vent port on the override control valve. Attention shall be given to the sequencing of the float valve supply and vent line porting so the system will not lockup (see 6.3.8) at any float position.

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3.11.4 Air-operated vent. The air-operated vent shall be spring-loaded shut to prevent the spillage of fuel and vapors under all conditions, including overturn. The vent shall be mounted in the top of the tank and as near the center as possible. The vent shall be capable of exerting a force of at least 150 pounds in order to break the vent loose under icing conditions. The vent shall have sufficient capacity to relieve a 750-gpm liquid or vapor flow without allowing the internal cargo tank pressure to exceed 2 psi. A bonnet shall be provided to prevent the entrance of rain, snow, and dust to the maximum practical extent when the vent is open and to prevent accumulation of snow and ice on the top of the vent. The bonnet shall also serve to deflect the fuel stream down onto the tank in the event of overfill.

3.11.4.1 Vent power system. The vent shall incorporate a double chamber actuating cylinder so that air in either chamber will open the vent. One chamber shall be used in conjunction with the bottom loading system and the other in conjunction with the pumping system. The actuating cylinders shall be designed fail-safe so that air cannot pass downstream to actuate the remaining portion of the bottom loading system or the pumping system interlock unless the vent is completely opened.

3.11.4.2 Pumping system interlock. An interlock shall be provided in the pumping system or truck chassis that will prevent fuel from being pumped into and out of the pumping system tank unless the vent is opened. The interlock shall also automatically shut down the pumping system in the event the vent closes. The interlock shall function without allowing the internal tank pressure to exceed -1 or +2 psig. The auxiliary vents shall not be considered in meeting this requirement. The interlock shall be incorporated into the pumping system hydraulic control system.

3.11.4.3 Vent controls. The vent shall incorporate two controls. The first control shall be located on or adjacent to the pumping system control panel. In the event the pumping system interlock requires opening the vent prior to shifting the power takeoff or accomplishment of some other function which the driver would perform prior to leaving the cab, the vent control shall be located in a convenient location in the truck cab. The first control shall allow release of the pumping system interlock only. The second control shall be located adjacent to the bottom loading inlet manifold and shall open the vent and then allow only the actuation of the remaining portion of the bottom loading system. The valve controls shall be of the lever-operated, on-off type. The system shall be so designed that both control valves can be actuated for simultaneous operation of the pumping system and bottom loading system. The first vent control, pump system interlock, shall include provisions for being held in the closed position by a padlock in accordance with MS35647-8 (the padlock is not to be provided).

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3.11.5 Control mounting. The bottom loading system vent control, level control tester, level control manual override, and manual control for the control valve shall be mounted on a control panel adjacent to the inlet manifold. The operating instruction and diagram plates shall also be mounted on or adjacent to the control panel.

3.12 Lights. The truck shall be equipped with the following lights to facilitate nighttime servicing:

3.12.1 Pumping compartment lights. At least two lights shall be provided in the pumping compartment. The lights shall illuminate the entire interior of the compartment and allow observation of all controls and instruments. The lights shall be vaporproof and shall incorporate threaded conduit connections.

3.12.2 Night servicing lights. The truck shall be provided with at least two fully adjustable lights to facilitate night servicing. Both lights shall have switches on the adjusting handle. The night servicing lights shall not operate unless the driving lights or parking lights are turned on. The lights shall be located as follows:

a. One light shall be a standard automotive spotlight and mounted in the forward left corner or driver's side of the cab roof. The light shall incorporate an ANSI Standard No. 4435 lamp. The light shall be so installed that it can be adjusted by the driver, but does not interfere with other normal driving functions or entering or exiting of the cab. If necessary for the truck to fit aboard cargo aircraft, a portion of the light above the cab shall be removable by one man with common handtools within 5 minutes. A weathertight quick-disconnect electrical connector shall be provided if it is necessary to remove the light for air transport.

b. The second light shall be located near the top left rear corner of the tank. The light shall be at least 8 feet from the ground. The light shall be adjustable by an operator standing on the ground. The light shall be so mounted that it will illuminate the rear or entire left side of the truck. The light shall incorporate an ANSI Standard No. 4412 lamp. The light assembly need not be vaporproof, however, the wiring to the light shall conform to 3.7.16.

3.13 Fire extinguishers. Two Type D-2 bromochloromethane fire extinguishers conforming to MIL-E-7589 shall be installed in their brackets. One fire extinguisher shall be mounted vertically on each side of the truck. Protection shall be provided from the splashing of mud and water. The fire extinguishers shall be readily accessible to personnel standing on the ground and shall not be mounted so they are behind the open pumping compartment doors.

3.14 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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* 3.15 Electromagnetic interference. The truck shall conform to the requirements of MIL-STD-461 as specified for class D3 equipment, method RE05.

3.16 Noise levels. The noise of the equipment shall not exceed the values specified in table I when measured at a 10-foot radial distance from the equipment operating at design capacity.

TABLE I. Noise Levels

Octave Bands (Hz)	Sound Level in Decibels Ref 0.0002 Microbars
20 to 75	100
75 to 150	100
150 to 300	100
300 to 600	90
600 to 1,200	90
1,200 to 2,400	90
2,400 to 4,800	90
4,800 to 10,000	85

3.17 Screw threads. Screw threads shall conform to Handbook H28.

3.18 Dimensions. The overall dimensions specified below shall be further reduced as required based upon the exact configuration of the truck to be furnished and the air transportability requirements specified herein.

3.18.1 Width. The overall width of the truck shall not exceed 102 inches with all access openings closed.

3.18.2 Height. The overall height of the truck shall not exceed 105 inches in the empty condition, not including the removable boarding ladder section and the light on the cab (see 3.12.2a). (For cab height - see 3.7.10.4).

3.18.3 Length. The overall length of the truck shall not exceed 34 feet.

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3.19 Finishes and protective coatings

3.19.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. A listing of the procedures used shall be furnished the procuring activity and shall be provided with the report specified in 4.4.2. Lacquer conforming to MIL-L-19537 may be used provided an acceptable procedure is approved by the procuring activity.

3.19.2 Painting. Exposed metal surfaces, including all surfaces inside the pumping compartment, shall be full gloss yellow, color code 13538, in accordance with MIL-STD-808. The chassis frame and running gear may be the manufacturer's standard black finish in lieu of yellow.

3.19.3 Anodizing. All aluminum moving parts and parts subject to wear shall be given an anodic coating.

3.20 Operational markings

3.20.1 Lettering and numbering. All lettering and numbering shall be reflectorized tape as specified in MIL-STD-808 and by the procuring activity (see 6.2).

3.20.2 Data plates. Instruction and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. The instruction plates shall be permanently attached and located on the inside of the compartment doors so as to be most accessible and visible to the operator standing in front of the control panel. The instruction and other data plates shall be aluminum and not less than 0.050 inch thick. Component identification plates shall be aluminum and not less than 0.032 inch thick. All plates shall be anodized or etched to produce raised natural metallic colored markings with a dark color background.

3.20.2.1 Diagrams. Schematic diagrams of the piping, including the bottom loading system and electrical systems, shall be provided. Each valve, switch, et cetera, on the diagram shall be properly identified to correspond to the markings on like parts in the truck.

3.20.2.2 Operating instructions. Operating instructions for the truck shall be clearly and concisely stated with reference to the actual accessories giving the identification as indicated on the wiring and piping diagrams. Any operations which affect safety of operation of the unit shall be listed at the top of the plate under the title CAUTION which shall appear in larger print or otherwise be designed to stand out. Specific instructions shall be listed separately for each pumping system operating mode with the pumping system mode as the title for the various sections on the plate. The plate shall list each step of the operation of the truck from the time the operator stops the truck.

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3.20.2.3 Bottom loading plate. A concise step-by-step operating instructions plate for the bottom loading system shall be permanently mounted on or adjacent to the bottom loading control panel. Any operations which affect safety of operation shall be listed at the top of the plate under the title CAUTION which shall appear in large print or otherwise be designed to stand out.

3.20.2.4 Transportation data plate. A transportation data plate shall be mounted on the inside of a pumping compartment door on the left or driver's side of the truck. The data plate shall contain at least the following:

a. Side and rear silhouette views of the truck

b. Horizontal and vertical location of the center-of-gravity of the truck on the silhouette views

c. Shipping weight

d. Loading cubage

e. Overall weight, width, and length.

3.20.2.5 Identification plates. Valves, switches, levers, and controls used in the operation of the truck shall be properly identified by identification plates. The data plates shall incorporate a numerical or alphabetical system of identification and an indication of the relative position of the lever or control handle such as on, off, 12V, et cetera. The plates shall be permanently fastened and conspicuously located on the referenced assembly.

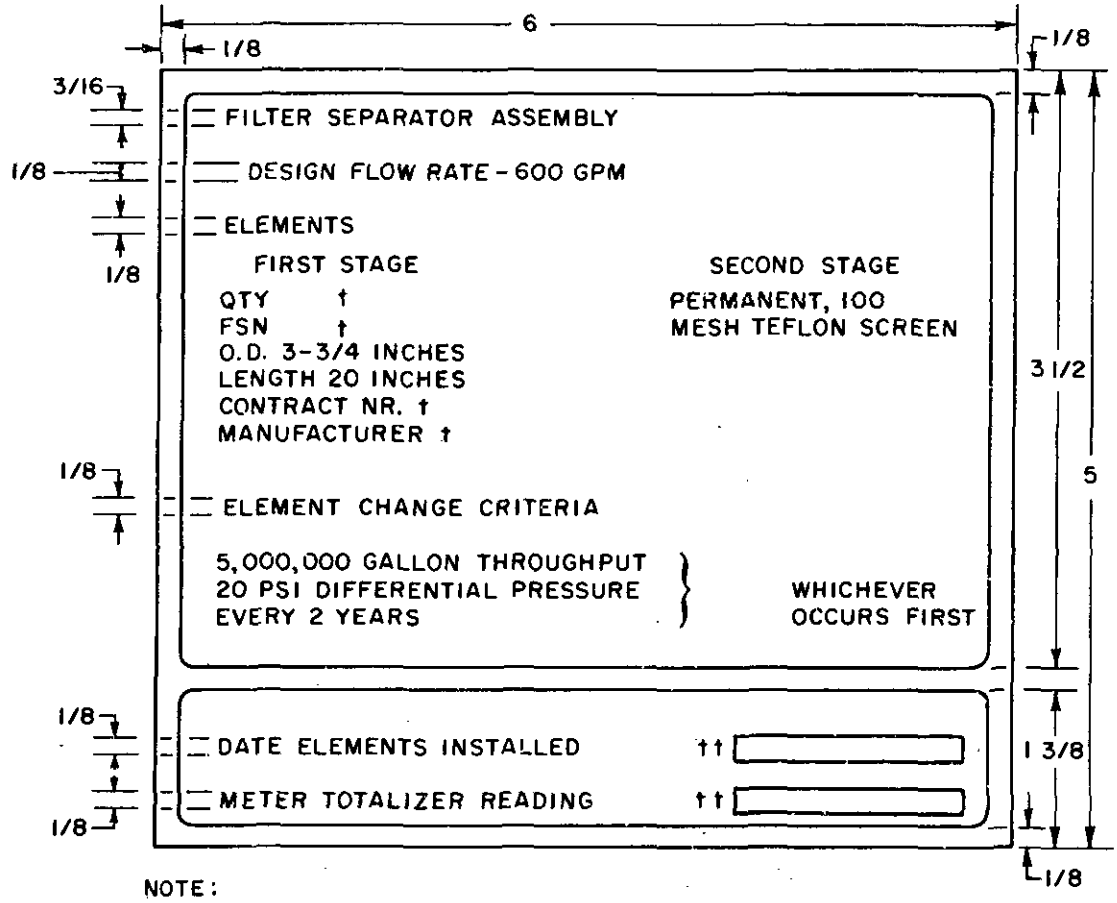
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3.20.2.6 Preproduction plates. Unless otherwise specified by the procuring activity (see 6.2), the preproduction trucks may be equipped with data plates consisting of typed lettering on paper laminated in plastic or similar semipermanent construction. The preproduction trucks shall contain sufficient identification and other data plates to provide examples of plates to be furnished on production trucks. The composition of all plates shall require the specific approval of the procuring activity. The preproduction trucks shall incorporate the final production model plates prior to delivery.

3.20.2.7 Filter-separator identification and instructional plate. A data plate as shown on figure 1 shall be mounted on the inside of one of the compartment doors on the left side of the truck.

3.20.3 Marking of tiedown points. The location and capacity of each tiedown point specified in 3.5.8.1 shall be clearly marked.

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NOTE:

1. †-APPLICABLE INFORMATION TO BE PROVIDED BY CONTRACTOR
2. ††-PROVIDE APPLICABLE SURFACE TO ALLOW DATA TO BE MARKED BY GOVERNMENT PERSONNEL
3. ALL LETTERING TO BE 3/32-INCH HIGH UNLESS OTHERWISE NOTED
4. BACKGROUND SHALL BE DARK
5. MATERIAL 16 GA. ALUMINUM
6. TOLERANCES ARE ± 1/32

FIGURE 1. Filter-Separator Identification and Instructional Plate

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3.20.4 The control lever specified in 3.7.4.1 shall be permanently marked to indicate the pump and drive positions.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. In addition, the acceptance date and gross weight shall be included.

3.22 Workmanship. The truck, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts, and tightness of assembly screws and bolts; et cetera.

3.22.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the truck, they shall be held or limited accordingly.

3.22.2 Screw assemblies. Unless torque is specified, assembly screws and bolts shall be tight.

3.22.3 Riveting. Riveting operations shall be carefully performed to insure tight and satisfactory heading of rivets.

3.22.4 Gear and lever systems. Gear and lever systems shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, and other irregularities. Where required for accurate adjustments, gear assemblies shall be free of backlash.

3.22.5 Cleaning. The truck and particularly the tank assembly shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slugs; stray bolts and nuts; pipe compound; rust; metal particles; and other foreign matter shall be removed during and after final assembly.

3.22.5.1 Weld beads. Weld beads shall be thoroughly cleaned, and spattering shall be removed from the tank.

3.22.6 Piping. The installation of piping, tubing, wiring, et cetera, shall be accomplished in such a manner as to prevent rubbing and wear due to vibration. The tubing and plumbing shall be assembled to permit adequate flexibility under torsional forces induced during operation. Failure in the piping shall not occur as a result of road operation or torsional forces.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Lubricants. Lubricants used in the truck during the tests specified under 4.6 shall be as recommended by the truck chassis manufacturer and the contractor for the mounted equipment and shall conform to existing military specifications for the type service.

* 4.3.2 Test fluids. Unless otherwise specified herein or in the detail component specifications, the test fluid used in the cargo tank shall be grade JP-4 turbine fuel conforming to MIL-T-5624 with a specific gravity of 0.76 or greater and the test fluid used in the engine shall be DF-2 diesel fuel oil in accordance with VV-F-800 and grade JP-4 fuel in accordance with MIL-T-5624 with the cetane rating included in the test report. Grade JP-5 turbine fuel conforming to MIL-T-5624 may be used for the hydrostatic pressure tests and for the pumping system tests in lieu of grade JP-4.

4.3.3 Data. During the testing specified herein, at least the following data shall be recorded at intervals not greater than 30 minutes, except when additional data and shorter time periods are a test requirement:

a. Date

b. Time started

c. Time finished

d. Ambient temperature

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e. Fuel level in cargo tank

f. Tank fuel temperature

g. Pressure at pump inlet (in Hg)

h. Pump discharge pressure

i. Filter outlet pressure

j. Fuel flow rate through each meter

k. Single-point nozzle pressure

l. Single-point nozzle pressure (from gage mounted on unit)

m. Engine speed (rpm)

n. Engine oil temperature

o. Engine water temperature.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. Unless otherwise specified by the procuring activity (see 6.2), one truck shall be subjected to the preproduction tests specified under 4.6 with exception of 4.6.26.7. A second truck shall be subjected to the preproduction test specified in 4.6.26.7.

4.4.2 Test report. Upon completion of the preproduction tests, a preproduction test report shall be prepared in accordance with MIL-STD-831. An authorized government quality control representative shall countersign all data sheets and the report. The following shall be incorporated as a part of an enclosure to the test report:

* a. 35-mm colored slides - left-front, three-quarter view and right-rear, three-quarter view as well as other photographs of the unit with test apparatus installed

b. List by all pumping system component manufacturers (major and minor components) of metals used in fabrication of respective components

c. Certification by all pumping system component manufacturers (major and minor components) that elastomers used in respective components are compatible with fuels conforming to MIL-T-5624 and MIL-G-5572

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d. A listing of all cleaning and finishing procedures in accordance with MIL-STD-808 to be used on the production trucks and all components

e. Report on all components covered by detail specifications requiring pre-production testing on the component

f. Chassis gradeability calculations in accordance with MIL-T-45341.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an attachment accompanying the preproduction test report (same quantity and quality of copies) or shall be incorporated as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Periodic sampling plan and tests . . . . . See 4.5.2

4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Mechanical operation . . . . . See 4.6.2

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c. Static pressure test . . . . . See 4.6.3

d. Pumping system hydrostatic tests . . . . . See 4.6.4

e. Inlet manifold hydrostatic pressure test . . . . . See 4.6.18.4

f. Hose reel tests . . . . . See 4.6.10

g. Pressure control tests - Contractor shall establish tests which assure all flow and pressure controls are properly set and function correctly.

4.5.2 Periodic sampling plan and tests. One truck shall be selected at random from the first 10 manufactured on a production contract, thereafter, one truck shall be selected at random from every 50 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Pumping system flow tests . . . . . See 4.6.5

b. Hose reel tests . . . . . See 4.6.10

c. Road handling test . . . . . See 4.6.26.1

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.1.1 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.2.1.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The truck shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to compliance with design requirements on preproduction trucks. Particular

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attention shall be given to materials, workmanship, dimensions, protective finishes and their application; welding, riveting; and markings on all trucks. For the preproduction truck, this examination shall be accomplished using a checklist which lists each requirement not validated by tests and the results of the examination after each requirement. As appropriate, the information as required by 4.4.2b through 4.4.2e shall be attached to the test report.

4.6.2 Mechanical operation. A thorough check shall be made of the truck to insure the proper functioning of all mechanical parts, accessibility of all controls and levers, interchangeability of removable parts, installation of equipment without fittings, and to effect every degree of safety and performance.

4.6.3 Static pressure test. The truck tank shall be subjected to an internal static air pressure of 3 psig for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tank to facilitate detection of leaks that may exist. Evidence of leakage shall be cause for rejection.

4.6.4 Pumping system hydrostatic tests. The flow system and components of the pumping system on the pressure side of the pump shall be subjected to a 30-minute hydrostatic test of 1-1/2 times the maximum total pressure encountered during operation of the unit or 200 psi whichever is greater. There shall be no evidence of leakage.

4.6.5 Pumping system flow tests. All pumping system flow tests shall be accomplished without any adjustment to the pumping system other than manipulation of controls by the operator in accordance with the operating instructions, except as specifically required herein.

4.6.5.1 Flow through overwing servicing nozzles test. The two overwing servicing hoses with nozzles installed shall be elevated to recirculate into the manhole of the cargo tank and the pumping system controls set for high-flow overwing operation. A flow rate of at least 300 gpm through each hose individually and at least 600 gpm simultaneously shall be demonstrated. Each condition shall be maintained for 5 minutes. By controlling only the overwing nozzles, flow rates of 0 and approximately 50, 100, 200, and 300 gpm through each hose individually and simultaneously shall be demonstrated without adjustment or manipulation of the pumping system controls. There shall be no instability in the pumping system operation. The pumping system shall demonstrate rapid flow recovery when the nozzle valves are opened after complete shutoff.

4.6.5.1.1 Overwing flow limiting test. With the maximum throttle setting and the nozzle valves held all the way open and the controls adjusted as specified in 4.6.5.1, it shall be demonstrated that the flow will not exceed 350 gpm through each hose either individually or simultaneously.

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4.6.5.1.2 Overwing low flow test. The two servicing hoses with nozzles installed shall be elevated to recirculate into the manhole of the cargo tank and the pumping system controls set for low flow overwing operation. A flow rate of 60 ±10 gpm through each hose individually and each hose simultaneously shall be demonstrated for 5 minutes with the nozzle valves held in the wide-open position. It shall be demonstrated that the flow through each hose can be controlled by the nozzles for any flow rate from 60 ±10 gpm down to zero both individually and simultaneously. The pumping system shall demonstrate rapid flow recovery when the nozzle valves are opened after complete shutoff.

4.6.5.2 Single-point flow tests. Single-point flow tests shall be conducted by connecting the Type D-1 nozzle to a mating receptacle connected to a length of pipe. A valve shall be installed downstream of the receptacle and used to induce backpressure on the single-point nozzle. Single-point nozzle pressure shall be measured downstream of the coupling strainer assembly in the upstream side of the single-point nozzle. The strainer shall be installed for all tests.

4.6.5.2.1 Maximum flow and pressure regulation test. The selector shall be set for high-flow single-point and the engine throttle adjusted for high flow operation. Once flow has been initiated, the gate valve shall be adjusted to provide a single-point nozzle pressure of 50 ±5 psi. The flow rate shall be at least 600 gpm. Without adjusting or manipulating any controls, the gate valve shall be adjusted to provide flows of 500, 400, 300, 200, 100 gpm, and no-flow. Each condition shall be maintained for at least 1 minute. The single-point nozzle pressure shall be 50 ±5 psi at each flow condition. Without adjusting or manipulating any controls, the gate valve shall then be adjusted from 0 to 600 gpm in 100-gpm increments as above. There shall be no instability in the pumping system operation. Data shall be recorded for each flow condition. Left and right meter readings shall be listed separately and shall in no case exceed 350 gpm.

4.6.5.2.2 Single-point flow limiting test. With the test setup specified in 4.6.5.2.1, the selector valve shall be set for high-flow single-point operation and the throttle advanced to the maximum. The gate valve shall be opened completely to provide a single-point nozzle pressure of approximately 10 psi. The flow rate through each meter shall be recorded and shall not exceed 350 gpm.

4.6.5.2.3 Surge pressure test. The unit shall be set for the 600-gpm, 50 ±5 psi condition as specified in 4.6.5.2.1. A quick-closing valve shall be installed downstream of the single-point receptacle. The valve shall close completely in 0.5 second. The valve shall incorporate a linear potentiometer to record the valve closure in a continuous trace on the recording paper along with the pressure. A pressure transducer of the strain-gage type with a response in the order of 1 ms shall be installed at the inlet to the single-point nozzle. The

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transducer pressure shall be recorded in a continuous trace using a recording oscillograph of the galvanometer light beam type. The amplifier and other electronic equipment shall provide for transient response in the order of 1 ms corresponding to the transducer. The quick-closing valve shall be actuated to shut off the flow in no more than 0.5 second. The valve closure trace and transducer pressure trace shall be recorded approximately 5 seconds prior to initiation of the closure to 10 seconds after the closure or when the nozzle pressure has stabilized, whichever is greater. The total pressure at the single-point nozzle shall not exceed 120 psi during the closure and the locked-in pressure shall not exceed 60 psi within 15 seconds after complete closure of the quick-closing valve. After the closure, the quick-closing valve shall be opened rapidly. The flow shall recover to 600 gpm within 15 seconds. The procedure shall be repeated five times. There shall be no adjustment or manipulation of the controls on the truck during the time spanned by the five test runs. A second transducer shall be installed in the pumping system upstream of the hose reels. Pressure traces for the second transducer shall be recorded during the five test runs. The transducers shall be calibrated with a deadweight tester immediately prior to and immediately after the test. The calibration curves shall be appropriately annotated and furnished with the transducer recordings in the test report.

4.6.5.3 Defuel test. When the defuel kit is provided, the following test shall be accomplished. The 50-foot, 2-1/2-inch defuel hose on the hose reel shall be attached to a storage tank or the truck cargo tank using the single-point nozzle with the strainer installed. The selector shall be set for DEFUEL and the fuel shall flow from the suction hose reel into the truck cargo tank through the system tank return line. A flow rate of at least 200 gpm shall be demonstrated for at least 5 minutes. The level of fuel in the storage tank or the truck cargo tank shall be no higher than 6 feet above the truck pump at the beginning of the test.

4.6.5.4 Eductor system test. Fuel shall be pumped through the servicing hoses to insure the hoses are flooded. The ability of the hose evacuation system to evacuate 90 percent of the fuel in the flooded hoses within 1.5 minutes shall be demonstrated.

4.6.6 Closed discharge test. The pump shall be operated at the maximum throttle setting for 30 minutes against a closed discharge. The fuel temperature in the pump shall not exceed 160°F during this period with an initial cargo tank fuel temperature of at least 70°F. There shall be not more than 1,500 gallons of fuel in the cargo tank. The closed discharge condition shall be with the hose reel and all other valves closed, except the tank shutoff valve and the system bypass operating.

4.6.7 Dry operation test. The pump and pumping system shall be completely drained. Without introducing liquid into the pumping system, the pump shall be operated at maximum rated rpm for 5 minutes. There shall be no appreciable wear, damage, overheating, or other type failure as a result of this test.

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4.6.8 Pumping system malfunction safety test. With both dispensing hoses rolled on the hose reels, the pumping system controls shall be randomly manipulated to demonstrate that the hoses cannot be pressurized under any condition without completely unreeling the hoses. The 3-inch hose shall then be unreeled and the single-point nozzle attached to the pumping manifold as specified in 4.6.5.2.1. All controls, including the selector, shall be randomly adjusted with the gate valve in the test manifold closed. The gate valve shall then be opened to the maximum and with the throttle advanced to the maximum it shall be demonstrated that flow rate of 350 gpm cannot be exceeded with the selector set at high-flow overwing. All adjustable pilots on the bypass valve and in-line control valve(s) shall then be adjusted to cause maximum restriction on the bypass. The gate valve in the test manifold shall then be closed and all controls including the selector shall be randomly adjusted. These test runs shall not produce pressure which could cause damage, failure, or permanent distortion of any part of the pumping system and the single-point nozzle flow pressure shall in no case exceed 55 or 60 psi static. Each control position shall be recorded along with the appropriate pumping system data.

4.6.9 Pumping system endurance test. The pumping system shall be operated at rated flow, single-point (600 gpm 50 ±5 psi nozzle pressure) for at least 100 hours. The system shall be continuously operated for periods of not less than 6 hours throughout the 100-hour endurance test. Each 2 hours of running, the gate valve on the test manifold shall be closed to shut off flow not less than four times and then opened to the original position without changing any of the truck controls. In addition, during each 2 hours of running the truck engine shall be idled at no-flow for 5 minutes. Wear, part damage, failure, or leakage shall not be evident as a result of this test.

4.6.10 Hose reel tests. The following tests of MIL-H-38209 shall be conducted on the hose reels as installed in the truck. In addition, the hose reels shall be checked for smooth operation when the hose is being unwound and rewound.

a. Mechanical operation

b. Unwind force

c. Rewind rate

d. Rewind force.

* 4.6.11 Water slug test. The truck shall be set for the 600-gpm, 50 ±5 psi condition as specified in 4.6.5.2.1. Water shall be introduced through the suction stub at a rate of 5 percent or greater so as to exceed the automatic water discharge capacity. The operation of the automatic water discharge valve

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to open and discharge water and the closing of the water slug valve with full cutoff of flow shall be observed. The introduction of water shall then be stopped and the truck shall automatically start flowing again when the water level in the filter-separator has been lowered by the automatic water drain. The above operation shall be repeated. During the test, samples in accordance with MIL-F-8901 for free water detection (viewer kit, free water detection) shall be taken downstream of the filter-separator at the sampling device every 2 minutes, except when the flow has stopped. The free water results shall not exceed 5 ppm or it shall be considered a failure.

* 4.6.11.1 Float-actuated selector valve test. The truck shall be set for the 600 gpm, 50 ±5 psi condition as specified in 4.6.5.2.1 and the float shall be manually moved through its complete range demonstrating automatic water discharge and water slug condition.

4.6.12 Single-point nozzle pressure gage test. A test shall be conducted to determine the error between the single-point nozzle pressure gage on the control panel and the actual single-point nozzle pressure. The test shall be conducted using the setup specified in 4.6.5.2.1. Beginning at 600 gpm and 50 ±5 psi, the flow shall be decreased in 50-gpm increments down to no-flow and then increased in 50-gpm increments to 700 gpm or the maximum flow. The pressures measured by the single-point nozzle pressure gage at the venturi throat or other point in the pumping system and a test gage used to measure pressure at the inlet to the single-point nozzle shall be recorded at each point. The nozzle pressure control pilot shall then be adjusted to at least two other random positions and the test run repeated.

4.6.13 Pumping system maintenance control adjustment test. As a part of the preproduction test report, complete written instructions for adjustment of the pumping system beginning with the controls in any random adjustment shall be provided. The instructions shall include diagrams as required. The instructions and demonstration of the complete adjustment procedure shall provide for sequential adjustment of all controls so that when any control is adjusted, that function will not be affected by subsequent adjustment of other controls according to the instructions. The control adjustment demonstration shall be conducted with all adjustable controls initially in a neutral-out-of-adjustment position.

4.6.14 Flexible coupling hydrostatic tests. All flexible couplings installed in flow lines between the tank and pumping system shall be hydrostatically tested. Each coupling shall be bench-tested on a test fixture to simulate the piping on each side of the coupling. The coupling, attachment, and clamping arrangement shall be identical to that used on the truck. Flexible couplings on the suction side of the pump shall be tested at 50 psi or three times the maximum working pressure, whichever is greater, and 15 inches Hg vacuum. Flexible couplings in the bypass line, tank return lines, or on the pressure

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side of the pump shall be tested at 200 psi or three times the maximum working pressure, whichever is greater. Flexible couplings on the pressure side of the pump which are subjected to negative pressures during operation of the unit shall also be tested at 15 inches Hg vacuum. The couplings shall be subjected to 100 cycles of 0 to maximum pressure to 0 and 100 cycles of 0 to 15 inches Hg to 0, if applicable. There shall be no leaks or evidence of wear, damage, or impending failure as a result of this test. It will not be necessary to subject grooved-type couplings to this test.

4.6.15 Tank backflow provisions test. The tank shutoff valve shall be closed. The pumping system shall be drained. It shall be demonstrated that the contents of the tank will not flow back into the pumping system under any tank load condition. The test shall be conducted with a cargo tank fuel level of approximately 1 foot or just above the bypass line inlet to the tank, whichever is higher, and repeated with the tank approximately filled to capacity.

4.6.16 Hydrodynamic loading test. The cargo tank shall be filled through the filler cover at a flow rate of at least 600 gpm through a 4-inch diameter pipe. There shall be no evidence of damage, failure, or permanent distortion due to the high flow rate stream. There shall be no excessive splashing due to interfering structure or other characteristics of the design of the tank.

4.6.17 Auxiliary vent operation test. The auxiliary vent operation test shall be conducted after the tank has successfully accomplished the test specified in 4.6.3. One of the two vents, or 50 percent of the venting capacity if more than two vents are provided, shall be removed from the tank. All tank openings, except the vent and manometer pressure taps, shall be effectively sealed. Fuel shall be pumped into and out of the tank at 10 to 20 gpm. The buildup of pressure and vacuum to plus and minus 0.2 psig, respectively, before the vents open shall be observed. Tank pressures shall be recorded at 15-second intervals, beginning when flow is started into or out of the tank and continuing until the plus and minus pressures in the tank stabilize. Fuel shall be pumped out of the tank at 50 gpm. The maximum negative pressure shall be recorded and shall not exceed 1 psig. Fuel shall be pumped into the tank at 50 gpm. The maximum positive pressure shall not exceed 2 psig.

* 4.6.17.1 Leak test. The venting equipment shall be removed from the tank and installed in a bench test fixture. The fixture shall be subjected to an internal pressure of 5 feet of water and oriented with the axis of the vent horizontal and then vertically downward. The vent leakage rate shall not exceed 40 milliliters per second in either position.

4.6.18 Bottom loading system tests

4.6.18.1 Static pressure test. The air portion of the bottom loading system shall be subjected for 10 minutes to a static air pressure of four times the maximum encountered during operation of the truck. Evidence of leakage or part failure shall be cause for rejection.

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4.6.18.2 Air system safety provisions test. A vent to atmosphere test valve shall be installed in the bottom loading pneumatic system. After the system has been fully pressurized, the valve shall be used to simulate a failure in the pneumatic system and release the air in the system. It shall be demonstrated that air will not be released from the air brake system to an extent that ample air pressure is not available for stopping the vehicle.

4.6.18.3 Vent opening load test. It shall be demonstrated that the vent will exert 150 pounds force upon opening on normal truck air pressure.

4.6.18.4 Inlet manifold hydrostatic pressure test. The inlet manifold assembly, including the control valve, shall be subjected to a 150-psi hydrostatic pressure for 10 minutes. There shall be no leaks in any portion of the manifold including the closed control valve.

4.6.18.5 Gravity discharge test. A single-point nozzle and a 10-foot length of 4-inch discharge hose or pipe shall be attached to the inlet receptacle. It shall be demonstrated that at least 4,000 gallons of fuel can be emptied from the tank by gravity at an average flow rate of 100 gpm.

4.6.18.6 Vent control test. The manual vent control shall be actuated five consecutive times while the tank is being filled at 500 to 600 gpm. It shall be demonstrated that the manifold control valve will not close in less than 3 seconds or more than 6 seconds.

4.6.18.7 Level control shutoff test. The tank shall be filled to capacity at a flow rate of 750 gpm. It shall be demonstrated that the manifold control valve will not close in less than 3 seconds or more than 20 seconds using the level control shutoff. The test run shall be repeated 10 times, and the level of fuel in the tank shall be $\pm$1 inch for all test runs. At least 1,500 gallons of fuel shall be discharged from the tank between test runs. The level control manual override shall be used to open the control valve initially when discharging fuel from the tank.

4.6.18.8 Level control tester. The tank shall be filled through the bottom loading manifold at 100 and 500 gpm a total of six times for each flow rate. During each filling operation, the level control tester shall be operated. The level control tester shall shut off the manifold control valve within 1 minute from the time the manual control is actuated during each tank filling.

4.6.18.9 Pumping system interlock test. Except for the power-operated vent and the manometer connection, all openings in the tank including the auxiliary vents shall be taped or otherwise effectively sealed. The pumping system shall be utilized to pump at 600 gpm and 50 $\pm$5 psi nozzle pressure into a storage tank other than the truck cargo tank. The pumping system shall then be used

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for defueling a storage tank other than the truck cargo tank at a flow rate of at least 200 gpm. Prior to each pumping operation, it shall be demonstrated the pumping system will not operate until the vent valve is opened. Prior to each pumping operation the vent valve shall be closed. It shall be demonstrated the negative pressure in the tank will not exceed 1 psi or the positive pressure of 2 psi. Each of the fuel-defuel operations shall be repeated 10 times. There shall be no malfunction throughout the entire test. There shall be no damage to any part of the pumping system.

4.6.18.10 Bottom loading system reliability test. The tank shall be filled and emptied at least down to the 1,000-gallon level a minimum of 30 times at approximately 750 gpm fill rate. At the beginning of each fill operation, the manual vent control valve shall be moved from the OFF to the ON position. The tank shall be filled until the level control closes the manifold control valve. The contents of the tank shall then be discharged using the level control manual override to open the manifold control valve. The cycle shall be repeated at least 30 times. Every fifth cycle, the level control tester shall be utilized. There shall be no evidence of malfunction of any component of the system during this test.

4.6.18.11 External load test of bottom loading receptacle. A single-point nozzle with a 100-pound weight attached to the nozzle inlet by a cable shall be connected to the bottom loading receptacle. The 100-pound weight shall be allowed to drop vertically a minimum of 4 inches, and shall be stopped by the cable. This dropping of the weight shall be repeated six times and shall not cause any visible damage or indicate any impending failure. The hydrostatic pressure test of 4.6.18.4 shall then be successfully accomplished.

4.6.19 Structural integrity and parts removal tests

4.6.19.1 Fenders. It shall be demonstrated that each fender will withstand a static load of 400 pounds placed on the top of the fender.

4.6.19.2 Pumping compartment roof loading. It shall be demonstrated that the pumping compartment roof as installed on the truck will withstand the weight of three men standing in any position without permanent distortion, damage, or evidence of lack of rigidity.

4.6.19.3 Pumping compartment roof removal. It shall be demonstrated that the pumping compartment roof can be removed by three men without tools in 3 minutes without damage to the roof or any portion of the truck.

4.6.19.4 Boarding ladder loading. It shall be demonstrated that a 220-pound man can scale the boarding ladder without evidence of deformation or lack of rigidity of the ladder or associated structure.

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4.6.19.5 Boarding ladder removal. It shall be demonstrated that one man with common handtools can remove the top portion of the boarding ladder as required for air transport in less than 5 minutes.

4.6.19.6 Defuel kit installation. When the defuel kit is provided, it shall be demonstrated that the defuel kit can be installed on the truck by two men using only handtools in 1/2 hour. There shall be no prior preparation of the truck.

* 4.6.20 Electromagnetic interference test. The truck shall be tested for radiated interference in accordance with Method RE05 of MIL-STD-461.

4.6.20.1 Electromagnetic interference tests need not be conducted on the hose reels since the entire truck will be tested as an end item.

4.6.21 Noise level test. The noise levels produced by the truck shall not exceed the values specified in 3.16 in any octave band when measured at a 10-foot radial distance from trucks operating at design capacity. Ambient background noise levels at all measurement positions shall be at least 10 db below the octave band sound pressure level produced by the trucks. The noise measurement instrumentation shall be as follows: The basic instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. This instrumentation shall meet the requirements specified in ANSI Standard S1.4 and Z24.10, except the octave band analyzer shall employ the preferred frequencies specified in ANSI Standard S1.6. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit the correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within $\pm$2 db. An acoustical calibrator shall be used for an end-to-end system calibration from the microphone to the meter. The sound pressure level (in db ref 0.0002 dynes/CM^2) produced by the calibrator when used in conjunction with the particular microphone in the system shall be known within $\pm$0.5 db.

4.6.22 Environmental tests. The following environmental tests shall be conducted on the truck in accordance with the specified methods of MIL-STD-810. Unless otherwise specified (see 6.2), an engineering evaluation may be performed in lieu of testing for the tests specified in 4.6.22.3 through 4.6.22.6. The engineering evaluation shall consist of a written detailed analysis of the truck assembly and all its parts and components to satisfactorily accomplish the required environmental conditions.

4.6.22.1 High temperature test. The truck shall be subjected to high temperature in accordance with method 501, procedure I. After being placed in the test chamber, the tests of 4.6.5.2.1 and 4.6.5.4 shall be conducted. The tank

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shall have at least 4,000 gallons of fuel. The temperature shall then be raised to 125°F which is the soak and operating temperature. At the end of the 48-hour soak period, the truck engine shall be started. The tests of 4.6.5.2.1 and 4.6.5.4 shall be conducted. The truck engine shall be stopped for 5 minutes and then restarted. The truck shall then be operated at rated capacity for 1/2 hour. The truck engine shall then be stopped for 5 minutes and restarted. The truck shall then be operated at idle speed with the pump engaged to the drive line and engine for 1/2 hour. The truck engine shall then be stopped for 5 minutes and restarted. All engine starts shall be accomplished within 5 minutes of the original attempt. In addition to the data required by 4.3.3, the following data shall be recorded during the operational portion of the test at 2-minute intervals:

a. Coolant temperature into engine

b. Coolant temperature out of engine

c. Engine oil temperature

d. Automatic transmission oil temperature

The truck shall operate satisfactorily to perform its intended functions throughout this test. There shall be no leakage, damage, evidence of thermal degradation, or impending failure of any portion of the truck. The engine and transmission oil temperatures shall not exceed the values recommended by the chassis manufacturer which shall be specified prior to conducting the test. The engine coolant temperature shall not exceed that recommended by the manufacturer or 10 degrees below the boiling point rating of the system, whichever is lower, and there shall be no loss of coolant. The truck engine shall be restarted without auxiliary cooling or electrical power.

* 4.6.22.2 Low temperature test. The truck shall be subjected to low temperature in accordance with method 502, procedure I, except the unit shall be cold-soaked for 48 hours at -65°F and then operated and inspected with the temperature maintained at -65°F. The truck shall be filled with 4,000 to 5,000 gallons of fuel at a temperature of not more than 80°F prior to the soak period. The engine fuel shall be grade JP-4 turbine fuel in accordance with MIL-T-5624 or DF-A diesel fuel oil in accordance with VV-F-800. The truck shall start within 20 minutes of the initial attempt without auxiliary heating or electrical power. The normal chassis winterization 110-V system may be operated as required throughout the test including the soak period. After the engine start and warmup, the tests of 4.6.5.2.1 and 4.6.5.4 shall be conducted, except the 600-gpm at 50 ±5 psi condition shall be maintained for at least 15 minutes. The bottom loading manifold shall be used to recirculate the fuel. All controls and components, including the bottom loading system, shall be operated to

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determine that malfunction will not result from the low temperature. The hose reels shall demonstrate proper operation and adequate power under the low temperature conditions. The truck shall operate satisfactorily to perform its intended functions throughout the entire test. The truck cab shall be tilted to the full-open position and then lowered to the normal position. The truck shall be inspected during the test and there shall be no leakage, damage, evidence of degradation, failure, or impending failure of any portion of the truck.

4.6.22.2.1 Upon completion of the above tests, the temperature shall be raised to 0°F and maintained at that level for 48 hours and the chassis winterization 110-V system shall not be operated. At the end of the 48-hour period, the engine shall be started twice without the use of external starting aids such as additional batteries, et cetera. However, an integral ether system controlled by the driver from inside the cab and glo-plugs may be used.

4.6.22.3 Humidity test. The truck shall be subjected to humidity in accordance with method 507, procedure I.

4.6.22.4 Fungus test. The truck shall be subjected to fungus in accordance with method 508, procedure I.

4.6.22.5 Salt-fog test. The truck shall be subjected to salt-fog in accordance with method 509, procedure I.

4.6.22.6 Dust test. The truck shall be subjected to sand and dust in accordance with method 510, procedure I.

4.6.23 Air transportability test. A detailed engineering evaluation shall be conducted on the truck to determine compliance with the tests specified in MIL-A-8421 for air transport in Types C-130, C-133, and C-141 aircraft. The evaluation shall include all major components and their capability to withstand the 8g forward load as well as the g-loadings specified in MIL-A-8421. The evaluation shall include the tiedown fittings. The evaluation shall also include a dimensional analysis for the truck while loading aboard a Type C-130 aircraft.

4.6.24 Dimensional check. A layout drawing of the truck shall be submitted with all dimensions referenced herein measured on the actual vehicle while standing empty on a flat surface.

4.6.25 Weight check. The weight on each wheel of the truck shall be measured with the unit standing on a flat surface, first empty and then with the cargo tank loaded with fuel equivalent to the rated load of 5,000 gallons at 6.6 pounds

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per gallon and with the pumping system fully flooded as during normal operation. The load on each side of the bogie assembly will be acceptable in lieu of the individual wheels of the bogie.

4.6.26 Mobility tests

4.6.26.1 Road handling test. The truck shall be filled with approximately 3,500 gallons of fuel and driven not less than 35 miles over paved highway or similar improved roads and not less than 15 miles over graded gravel or similar unimproved roads. Where practical, the truck speed shall be 45 to 50 mph on improved roads and 25 to 30 mph on unimproved roads. At least 10 short-radius right turns, 10 short-radius left turns, and 5 sudden stops shall be conducted. Throughout the road handling test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, et cetera. Upon completion of the road handling test, the truck shall be inspected for misalignment, distortion, or other damage sustained as a result of the testing. Undesirable roadability features and any damage resulting from the test shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be recorded on the data sheets.

* 4.6.26.2 High speed tests. The truck shall be loaded to design capacity. It shall be demonstrated that the truck can achieve a speed of 50 mph on level, paved highway within 2.5 miles from a dead stop. It shall be demonstrated that the truck can achieve a speed of 40 mph on a level, graded gravel road within 1.5 miles from a dead stop. The tests shall be conducted using DF-2 diesel fuel oil as the truck engine fuel; and the tests shall be repeated using grade JP-4 turbine fuel in accordance with MIL-T-5624, except the speeds obtained shall be as specified in 3.6.7.

4.6.26.3 Service brakes. The truck shall be loaded to design capacity. The truck shall be subjected to three emergency stops in rapid succession from 20 mph on a clean, dry, smooth, level paved surface free from loose material. Each stop shall be accomplished in not more than 1-minute intervals. The truck shall stop within 30 feet from the point the brakes are applied. A time-distance recorder shall be used and shall positively identify the point where the brakes are initially applied.

* 4.6.26.4 Airbrake emergency test. The airbrake emergency brake system shall be demonstrated.

* 4.6.26.5 Ramp clearance test. The truck in the empty condition shall enter, negotiate, and leave a ramp with a slope of 15 degrees in both the forward and reverse directions. There shall be no contact between any part of the truck, except the wheels, with any part of the ramp surface including the horizontal top and bottom landings. The length of the ramp, measured along the slope,

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shall be greater than the wheelbase of the truck. The bottom and top of the ramp shall end in horizontal landings which form a sharply defined 15-degree angle with the ramp.

4.6.26.6 Suspension articulation test. The fully loaded truck shall be driven over 8-inch-high obstacles. Unless otherwise approved by the procuring activity, the obstacles shall be 8-inch high blocks, one for each track. The blocks shall first be side-by-side in each track and then at an angle of 45 degrees across the tracks to simulate the truck negotiating an object at 45 degrees to the path of the vehicle. The blocks may be sloped to facilitate driving over, however, the blocks will be short enough that when one wheel on the bogie is up 8 inches, the other wheel on that same side will be on the ground. There shall be no interference, part failure, or permanent or objectionable distortion of any part of the truck as a result of this test.

* 4.6.26.7 Five-thousand-mile test. The truck shall be driven 5,000 miles (see 6.4) consisting of five cycles of 1,000 miles each. The first cycle shall be run with the tank half filled with fuel. The second, fourth, and fifth cycles shall be run with the tank filled with fuel to rated capacity. The third cycle shall be run with the tank empty. The truck engine fuel for cycles 1 through 4 shall be DF-2 diesel fuel oil in accordance with VV-F-800 and for cycle 5 shall be grade JP-4 turbine fuel in accordance with MIL-T-5624. At the end of the second and fourth cycles, the pumping system shall be operated 1/2 hour at rated capacity, 600 gpm at 50 $\pm$5 psi. At the end of the fifth cycle and with the engine operating on fuel in accordance with VV-F-800, the pumping system shall be operated for 1/2 hour at rated capacity and the hydrostatic test of 4.6.4 shall be conducted. During the paved highway portion of each cycle, at least 25 short-radius right turns, 25 short-radius left turns, and 5 sudden stops shall be accomplished. The pumping system, including the filter-separator shall be filled with fuel throughout all five cycles. Each cycle shall consist of the following phases:

a. Highway - 750 miles at an average speed of at least 40 mph and a maximum speed of 50 mph.

b. Gravel road - 200 miles at an average speed of at least 25 mph and a maximum speed of 40 mph.

c. Belgian block pavement - 50 miles at an average speed of at least 10 mph and a maximum speed of 15 mph.

4.6.26.7.1 Servicing and adjustments. The truck shall be serviced and the brakes adjusted prior to the road test. Periodic servicing with lubricants and preventive maintenance adjustments shall be permitted during the testing at the intervals specified in the operation and service instruction handbook. Tires shall be serviced as necessary. Other servicing, reconditioning, or repair shall not be permitted without prior approval of the procuring activity.

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4.6.26.7.2 Inspection. Throughout the road endurance test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, et cetera. Upon completion of the road endurance test, the truck shall be inspected for excessive wear, misalignment, distortion, or any sort of structural failure or impending failure of any portion of the truck. There shall be no leaks or malfunction of the pumping system or tank. A complete description of any excessive wear, damage, failure, impending failure, or roadability deficiencies shall be recorded on the test data sheets.

4.6.26.8 Vehicle clearance circle test. The vehicle shall demonstrate the ability to meet the vehicle clearance circle requirements of 3.7 both clockwise and counterclockwise.

* 4.6.27 Reliability demonstration. Compliance with reliability requirements shall be considered demonstrated when the following conditions apply:

* a. All tests specified herein shall be completed with no reliability failure. The total accumulated operating time for all equipments under test shall be computed when the preproduction tests are completed. If this total, for all tests is less than 194 hours, additional operating tests shall be conducted, under the endurance tests, until that required time is accumulated.

* b. Any test resulting in a failure shall be documented. Apparent failures that are caused by some unintended factor external to the equipment (for example, test facility failure) need not be counted as reliability failures if the causes can be substantiated to the procuring activity's satisfaction.

* 4.6.28 Maintainability demonstration. Compliance with the maintainability requirements shall be demonstrated by inspection and examination of the truck from the standpoint of ease of maintenance and servicing. Particular attention shall be given to maintenance with a minimum number of common tools, and to servicing and operation without undue difficulty. Signs of excessive wear or other indications of potential failure shall be reported as to when and where observed, and the information supplied to the procuring activity accompanied by proposed corrective action.

4.6.28.1 Tools and equipment. Tools and equipment recommended for maintenance shall be used to the maximum practicable extent during testing with emphasis on any special tools or equipment required. Use of special tools and equipment shall be sufficient to determine that they are useful and justified. Instances where special tools or equipment are inadequate shall be recorded in detail.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Trucks for domestic destinations shall be prepared for shipment in accordance with the supplier's standard practice and shall be acceptable to the carrier at lowest rates. Trucks for overseas destination shall be prepared for shipment in accordance with level B (mobile) requirements of MIL-STD-281, except that the engine crankcase shall be filled to the operating level with the proper grade of type P-10 preservative oil and preservative shall not be applied to interior surfaces of the tank body and pumping system. Batteries shall be secured in place in the vehicle battery carriers.

5.1.1 Cooling system. Cooling systems shall contain a clean solution of equal parts by volume of antifreeze (ethylene glycol) conforming to Type I of O-A-548 and water filled to capacity. A warning tag bearing the information COOLING SYSTEM FILLED WITH WATER AND ETHYLENE GLYCOL IN EQUAL PARTS BY VOLUME - DO NOT DRAIN shall be securely attached to radiator filler neck.

6. NOTES

6.1 Intended use. The A/S32R-9 truck is intended for use in servicing fuel to all types of USAF aircraft under world-wide conditions.

6.2 *Ordering data. Procurement documents should specify the following:*

a. Title, number, and date of this specification

b. That MIL-M-008090E(USAF), dated 18 April 1966, or later issue be used in conjunction with this specification (see 3.7.5)

c. When a pintle hook is to be provided with the production truck (see 3.7.11)

d. When 1-1/2-inch ID hose and a Type MD-1 nozzle is to be provided with the production truck (see 3.10.6)

e. When a Type MD-2 nozzle with a quick-disconnect coupling is to be provided for the 3-inch hose (see 3.10.6.1)

f. The quantity required, and when the defuel kit is to be installed (see 3.10.12)

g. Lettering and numbering required on the truck (see 3.20.1)

h. If preproduction trucks cannot be equipped with data plates consisting of typed lettering on paper laminated in plastic or similar semipermanent construction (see 3.20.2.6)

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i. Location and conditions for preproduction testing (see 4.4)

j. When more than one truck is to be subjected to the tests specified under 4.6, except 4.6.26.7 (see 4.4.1)

k. When an engineering evaluation will not be accepted in lieu of the testing specified in 4.6.22.3 through 4.6.22.6 (see 4.6.22).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Mean-time-between-failure. Mean-time-between-failure is defined as the arithmetic mean (average) operating time between failures that could prevent satisfactory or safe performance. Note that accumulated test time equal to the specified MTBF does not demonstrate that MTBF at stated confidence. The full number of test cycles or test time specified in section 4 are required.

6.3.2 Failure. For reliability purposes, a failure shall be defined as any failure or malfunction that could prevent safe operation or that could prevent satisfactory performance, within limits specified herein, in the primary functions of loading, storing, transporting, and delivering fuel. Malfunctions that would not prevent such primary performance, such as loose bolts in non-critical applications, need not be counted as reliability failures.

6.3.3 Self-priming centrifugal pump. A self-priming centrifugal pump is one designed to have the ability to prime itself automatically, after being initially filled, when operating under a suction lift, and to free itself of entrained gas without losing its prime and to continue normal pumping without attention.

6.3.4 Highway. A highway is a road of level concrete or macadam.

6.3.5 Unimproved road. An unimproved road is a road consisting of level-to-rolling gravel trails.

6.3.6 Belgian block pavement. Belgian block pavement consists of a very rough, wavy, hard surface unaffected by changes in climatic conditions.

6.3.7 Hazardous vapor area. The hazardous vapor area is defined as the entire rear portion of the truck from the back of the cab.

6.3.8 Lockup. Lockup is defined as a condition when air cannot be supplied to the manifold valve cylinder by the override control and the normal supply due to the position of the float.

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6.4 Test courses. The test courses in existence at Aberdeen Proving Ground, Aberdeen, Maryland, on the date of invitation for bids will be considered the standard test courses. Should the tests be conducted at an alternate test course, the contractor is still responsible for designing and constructing the vehicle to successfully perform the tests on the standard test courses at Aberdeen Proving Ground.

6.5 International standardization agreements. Certain provisions of this specification, as specified below, are the subject of international standardization agreements as follows: Air Standardization Co-ordinating Committee Air Standard (ASCC Air Standard), North Atlantic Treaty Organization Standardization Agreement (STANAG), and Southeast Asia Treaty Organization Standardization Agreement (SEASTAG). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

MIL-T-83219B(USAF) Paragraph	ASCC AIR Standard And Paragraph	STANAG And Paragraph
3.6.2c	11/3C, (b)	3105, Ed. No. 2, 1a(1)
3.6.2d	11/3C, (c) (d)	3105, Ed. No. 2, 1g
3.10.3	11/6B 7, (d)	3334, *(See Note) 1(d)
3.10.4		3105, Ed. No. 2, 1c
3.10.6		3105, Ed. No. 2, 1d
3.10.6.2	11/3C, (a)	3105, Ed. No. 2, 1b(2)

NOTE: *Also applies to SEASTAG 3334, paragraph 1(d)

6.6 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

ac . . . . . alternating current
amp . . . . . ampere
db . . . . . decibel
g . . . . . gravitational unit
gpm . . . . . gallons per minute
GVW . . . . . gross vehicle weight
Hg . . . . . mercury
Hz . . . . . Hertz, cycles per second

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ID	inside diameter
ms	millisecond
mph	miles per hour
MTBF	mean time between failure
NPT	national taper pipe (thread)
psi	pounds per square inch
psig	pounds per square inch gage
PTO	power takeoff
rpm	revolutions per minute
V.	volt.

6.7 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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